How Far Can Gerry Mander? Redistricting with Endogenous Candidates*

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Abstract

I study partisan gerrymandering when district composition affects candidates' policy positions and voters' behavior. In the U.S., primary elections determine competing candidates, with districts' ideological composition shaping nominee selection. Thus, redistricting affects not only which party wins but also candidates' ideologies. I find that classical gerrymandering strategies backfire when candidates emerge endogenously, particularly where extreme voters may select non-viable candidates. However, when properly designed to account for voter affiliation and preference intensity, gerrymandering becomes more powerful than traditional approaches. Using optimal transport theory, I characterize the optimal redistricting plan, which maximizes ideological distance across opposition voters in the same district. I analyze implications for polarization and minority representation.

Keywords: Gerrymandering, optimal transport, information design, polarization.

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1 Introduction

Partisan gerrymandering is the deliberate manipulation of electoral district boundaries to favor a political party. The 2012 Republican Redistricting Majority Project (REDMAP) is an example: despite receiving 1.4 million fewer votes than Democrats in U.S. House elections, Republicans secured a 33-seat majority through redistricting (Daley, 2020). Traditional approaches like REDMAP rely on presidential election data to classify areas as Republican or Democratic, implicitly assuming voter behavior remains unaffected by redistricting itself. The 2022 House elections offer an intriguing counterpoint: Republicans underperformed expectations, losing competitive districts they were favored to win, often where extreme candidates had emerged from primaries (Thakker, 2022; Rakich, 2022; Korte et al., 2022).

This paper studies optimal redistricting when candidates' policy positions and, consequently, voters' behavior, respond to the districts in place. I embed a two-stage model of elections at the district level into an optimal redistricting problem and illustrate how traditional gerrymandering can backfire when extreme voters select non-viable primary candidates. Using methods from optimal transport theory (Monge, 1781; Kantorovich, 1942), I show how savvy redistricters can create more powerful plans than standard redistricting allows for. Their strategy? Crafting district boundaries that drive a wedge between moderates and extremists in the opposing party, effectively turning their rivals' diversity into a liability.[1]

Illustrative examples. To illustrate the forces in my model, it is useful to recall the standard redistricting setting. In the example in Figure 1, there's a U.S. state with two-thirds Democrats (in blue) and one-third Republicans (in red). A Republican redistricter has to partition voters into, say, three equipopulous districts, and wants to maximize the number of districts with a Republican majority. She applies the standard "pack-and-crack" technique. That is, she creates two "cracked" districts, District 1 and 2, consisting of just enough Republicans to have a majority, and one "packed" district, made up of only Democrats. Thanks to gerrymandering, Republicans win two-thirds

[1] Redistricting can affect voter behavior through multiple channels. I focus on voters switching party when they dislike their nominee, but the key insight—that optimal gerrymandering exploits wedges between voter types—extends naturally to other mechanisms, like endogenous turnout. See more in Section 5.2.

of districts with just one-third of overall Republican supporters. This is a classic result and one core point of the effects of gerrymandering.

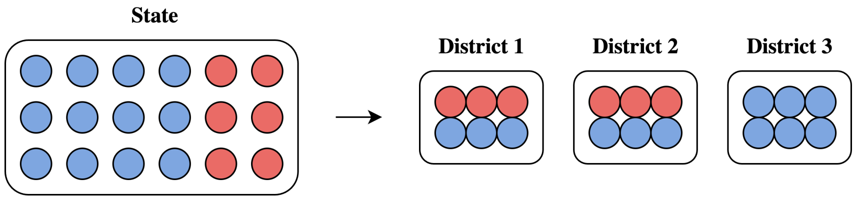


Figure 1: Standard pack-and-crack redistricting.

The approach above, however, assumes that party positions are fixed and exogenous. Suppose that party positions in each district are endogenous. Consider Figure 2. In the population, a small fraction of Democrats and an even smaller fraction of Republicans are moderates, represented by less intense shades of blue and red, respectively; all other voters are partisans (or extremists). Suppose primary elections precede the state-wide elections. In each district and for each party, primaries determine the candidate, who can be either a moderate or an extremist, depending on which voter type constitutes the majority within their party. In general elections, while partisans consistently vote for their party's candidate, moderate voters prefer a moderate candidate from *either* party over an extreme candidate from their own party. The standard "pack-and-crack" gerrymandering strategy can backfire. For example, consider District 1 and District 2 in Figure 2. In both districts, there is a majority of partisans among Republicans and a majority of moderates among Democrats. Hence, the Republican candidate caters to partisans, while the Democratic candidate caters to moderates. The only moderate Republican voter prefers voting for the moderate Democratic candidate rather than for a partisan Republican. Redistricting effectively converts the moderate Republican voter to a Democratic voter! Ultimately, with endogenous candidates, classical gerrymandering fails to achieve its goal: in this example, *all* districts end up selecting a Democratic candidate.

A real-life equivalent of such "dummymandering", as a bad gerrymandering is usually referred to in policy circles, can be found in the case of Oregon's fifth district: after Democrats redrew boundaries in anticipation of the 2022 mid-term elections, progressive Jamie McLeod-Skinner unexpectedly defeated seven-term centrist incumbent Kurt Schrader in the Democratic primary. Crucially, McLeod-Skinner's victory hinged

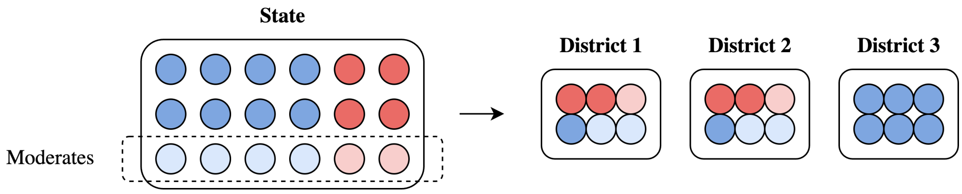


Figure 2: Pack-and-crack backfires.

on a forty-point advantage in Deschutes County, which was only added to the district through a recent redistricting effort. This shift in Democratic candidate allowed Republican Lori Chavez-De Remer to appeal to moderate voters and flip the district for the first time since 1994 (Flaccus, 2022; Glueck, 2022; Scott and Weigel, 2022).[2]

Do such examples and real-life instances mean that gerrymandering is less powerful than previously thought? Quite the opposite. Consider a scenario where a Republican redistricter is aware of and exploits the endogeneity of voter behavior. Figure 3 illustrates an optimal gerrymandering plan. In District 1, despite having a Democratic majority, the composition is heterogeneous. There are sufficient Democratic partisans so that the Democratic candidate caters to their preferred policy, while the two moderate Democrats align more closely with the Republican candidate, who caters to the single moderate Republican voter in the district. Consequently, moderate Democrats prefer to vote Republican, even though they would have preferred a moderate Democratic candidate to a moderate Republican. As a result, Republicans win this district. District 3 consists of a majority of partisan Republicans, ensuring a straightforward Republican win. By strategically considering the endogeneity of candidates' selection and voters' behavior, the redistricter manages to secure *all districts* for the Republican party, despite Republicans constituting only one-third of the overall population. That is, gerrymandering can be more powerful than previously thought if its impact on voting behavior is taken into account.

Model. There is a continuum of voters with single-peaked preferences over a unidimensional policy space. Voters are identified by their bliss point, which I sometimes refer to as their "type." Republicans are voters with bliss points above an exogenous cutoff, while Democrats are voters with bliss points below the cutoff.

[2]Another example is Ohio's ninth district, which was predicted to have R+3 to R+6 partisan lean, but ended up being won by the incumbent Democratic candidate by a thirteen points advantage (Lavelle, 2023).

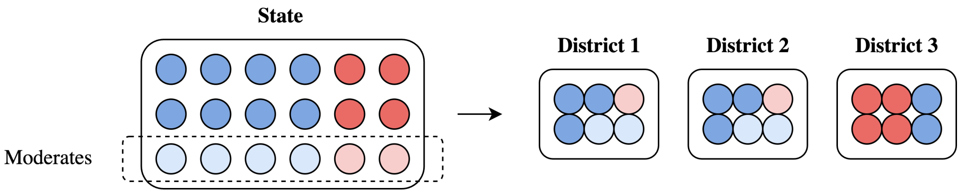


Figure 3: Gerrymandering is even more powerful than previously thought.

A Republican designer partitions voters into equipopulous districts to maximize the number of districts won by Republicans.

The model incorporates a two-stage election process at the district level to determine voting behavior. First, in the primary elections, Democratic and Republican voters separately select their party candidates. Then, in the general election, all district voters choose between the two primary winners. Drawing on the work of Owen and Grofman (2006), I model electoral incentives so that primary candidates position themselves to appeal to their respective party medians.[3] To close the model, in districts lacking Republican or Democratic voters, I assume the corresponding party's candidate defaults to the most moderate position within their ideological spectrum. The general election outcome is decided by the district's median voter. The candidate whose position most closely aligns with the district median wins the election.

Finally, I allow the designer to face some uncertainty about voters' preferences, which I parametrize by a one-dimensional aggregate shock. In particular, the case of a small aggregate shock is of special interest as it selects a unique, robust solution from the multiplicity that arises without uncertainty. Moreover, recent evidence (Kolotilin and Wolitzky, 2026) suggests it is the empirically relevant case.

Results. In Section 2, I show that the problem can be mapped to a Monge-Kantorovich optimal transport problem (Theorem 1), where the designer determines how to optimally "transport" voter types from above the population median to match with types below it.

The result hinges on Proposition 1, establishing that each optimal district must con-

[3] In the Supplemental Appendix, I provide microfoundations. The strict dependence on party medians is not necessary for my results: the driving force behind the model is the responsiveness of candidates to party preferences. For instance, the model can be extended to situations where candidates position themselves at a convex combination of party medians and the overall district median. In Section 5.1, I provide an extension where candidates' positions are determined by a general quantile of the preference distribution within each party.

tain exactly two voter types: one type from below the population median and one type from above it, in equal proportions. The intuition emerges from the examples above. First, Figure 2 demonstrates how pack-and-crack backfires when District 1 mixes moderate and extreme Republican types. The designer can perfectly hedge against this possibility by creating cracked districts with no more than one Republican type, ensuring the optimal plan performs at least as well as standard pack-and-crack. Second, Figure 3 reveals a novel opportunity: In Districts 1 and 2, the ideological divide between moderate and extreme Democrats allows Republicans to win despite being in the minority. This same mechanism extends to districts packed exclusively with Democrats, where a moderate Republican candidate can peel off enough moderate Democratic voters to secure victory. To maximize this advantage, each such district should contain exactly two voter types: a higher type serving as moderate Democrat and a lower type as extreme Democrat. This hybrid strategy—packing Democrats while cracking them between moderates and extremists—allows the optimal plan to strictly outperform traditional pack-and-crack.

The effectiveness of gerrymandering depends critically on voter type differences within a district. The larger the wedge between the two voter types, the higher the probability of winning the district, making districts with significant gaps "safer" than more homogeneous ones. While the designer would ideally maximize this gap for all districts, she is constrained by the overall distribution of voter types. She therefore has to decide which voter types to allocate to safer, more heterogeneous districts, and which to less safe, more homogeneous districts. The solution to this problem depends on the distribution of uncertainty about voter preferences. Whenever the cumulative distribution function (CDF) of the shock is neither concave nor convex, the optimal transport problem turns out to be non-standard.

In Section 3, I characterize the solution under the assumption that the aggregate shock is S-shaped and symmetric around zero.[4] Proposition 3 demonstrates that there is a unique optimal redistricting plan. Voters far from the population median are matched in a "positive assortative" or "comonotonic" manner: high types above the median are paired with high types below it. However, for voters close to the median,

[4]The CDF of the shock is convex below zero and concave above it. Prominent cases of symmetric noise considered in the literature—e.g., normally-distributed noise—satisfy this assumption.

the matching reverses to "negative assortative" or "anti-comonotonic": high types above the median are paired with low types below it.

The intuition is as follows. The districts most likely to be won are those with large ideological gaps between their two voter types, which requires matching voters from the right tail of the distribution with voters from the left tail. When working with these extreme voters, there's no benefit in creating an excessively large gap in one district at the expense of another. Instead, the designer uses positive assortative matching to maintain consistent, sufficient gaps across all such districts. However, closer to the median, ideological gaps become harder to create and districts become less likely to be won. Here, the designer switches to negative assortative matching: creating some very safe districts with large gaps at the expense of other districts that are given up as lost.

Further characteristics of the solution depend on the precise shape of the distribution of voters and of the shock. Of particular interest is the solution when the variance of the shock is small. In that case, the designer can reduce the mass of negatively assortative matches and redistribute it to positive assortative matches—where extreme types above the median pair with moderate types below it. In Proposition 4, I show that as the variance of the shock goes to zero, the solution converges to a distribution that allocates as much mass as possible to the region where positive assortative matches happen.

In Section 5.1, I extend the model to consider a scenario where candidates' positions are determined by a general quantile of the preference distribution within each party, rather than just the median. For the case of a uniform distribution of voter preferences, I show that the optimal plan involves a three-wise positive assortative matching. This result demonstrates that the strategic creation of wedges between voter types remains crucial even under alternative candidate positioning rules, though the specific implementation of the strategy adapts to the new context.

Implications. The model yields implications for the current political and legal debates. For instance, when the distribution of voters is sufficiently "spread-out", the Republican redistricter benefits by more easily exploiting the wedge between moderate Democrats and extreme Democrats (Proposition 5). In Section 4, I examine how optimal gerrymandering strategies can exacerbate polarization in the U.S. House of Representatives. Proposition 6 predicts an ideological gap among elected representatives, who are either moderate Republicans or extreme Democrats. Notably, extreme Democrats emerge in districts with a significant share of moderate voters, challenging the conventional view that political extremism stems from segregating extreme voters into homogeneous districts. Finally, I discuss the implications for minority representation in congressional districts. In "minority opportunity" districts, where minorities make up 40-50% of voters, minority candidates' success depends critically on white voter support. My model suggests this dependence can trigger "white backlash," where white voters unite against minority-preferred candidates, benefiting Republican candidates as predicted by my framework.

1.1 Related Literature

Optimal partisan gerrymandering. This paper primarily relates to the economic theory literature on optimal partisan gerrymandering, starting with Owen and Grofman (1988). While their work focuses on a binary voter type model, subsequent research has emphasized the importance of considering voters' preference intensities in redistricting strategies. In existing models, preference intensities play a role mostly due to uncertainty about voter preferences, typically assuming moderate voters are more susceptible to preference swings. Friedman and Holden (2008) show that with significant aggregate preference uncertainty, the optimal redistricting plan assigns extreme supporting voters to the same districts as extreme opposing voters, effectively neutralizing the latter's influence. The authors term this approach "matching slices," which is reminiscent of what I term negative assortative matching. Kolotilin and Wolitzky (2026) develop a more comprehensive model of gerrymandering that allows for both aggregate preference uncertainty and substantial shocks that affect voters independently, namely idiosyncratic uncertainty. Their findings suggest that when idiosyncratic uncertainty dominates, the optimal redistricting involves segregating the most extreme opposing voters while matching the remaining voters in a negative assortative manner. Competition between two parties, each controlling redistricting in distinct areas, is examined by Gul and Pesendorfer (2010). In their optimal plan, opposing voters are segregated, and more favorable voters are all pooled together.

I contribute to this literature by allowing individual voters' behavior to depend on the composition of their district — specifically, on the types and ideological positions of the other voters allocated to it. One implication of this approach is that it results in a different treatment of extreme opposing voters. While previous literature either predicts such extreme voters to be segregated or matched with extreme supporters, my model exploits them to turn moderate voters against their own party. To do so, the optimal plan prescribes "mis-matched slices," on either side of the preference distribution, resulting in novel implications.

Information design. As Kolotilin and Wolitzky (2026) show, the gerrymandering problem can be mapped onto an information design problem. The distribution of voter preferences serves as the "prior," districts function as "posteriors," and a redistricting plan represents a distribution of districts that satisfies a constraint, which is mathematically equivalent to Bayes plausibility. Indeed, in the special case of exogenous policies, my model becomes a variant of Bayesian persuasion (Kamenica and Gentzkow, 2011) for medians, which can be solved with recent off-the-shelf tools (Yang and Zentefis, 2024), as I show in Section 2. My solution sheds light on information design problems where payoffs depend on more intricate aspects of posterior distributions, such as the relative positions of conditional medians, rather than single summary statistics like means or medians.

Optimal transport. To characterize the solution, I leverage a Monge-Kantorovich optimal transport representation of the redistricter's problem (Monge, 1781; Kantorovich, 1942). Drawing on results from Chiappori et al. (2010) and Santambrogio (2015), I establish the existence and uniqueness of the solution and characterize it for the case of a surplus function that is symmetric and S-shaped, thus quite different from the typical functional forms in the optimal transport literature. The only other papers I am aware of that establish explicit properties of the solution for a concrete optimal transport problem with a non-typical surplus function are Strack and Yang (2025) and Boerma et al. (2025).

Other topics in gerrymandering. The broader literature on gerrymandering tackles a variety of different issues. The effects of redistricting on policy choice are considered by Shotts (2002) and Besley and Preston (2007), while the impact of gerrymandering on polarization in the House of Representatives is addressed, for instance, by McCarty et al. (2009). Other important topics in redistricting that I do not explore in this paper relate to: including geographic constraints on gerrymandering (Puppe and Tasnádi, 2009), accounting for differential voter turnout (Bouton et al., 2023), and measures of electoral maldistricting (Gomberg et al., 2023).

2 The model

2.1 Setup and Statement of the Problem

Voters and Parties. Consider a continuum of voters with single-peaked preferences over uni-dimensional policy space $[\underline{v}, \overline{v}]$, a closed interval on the real line $\mathbb{R}$. A voter's ideal point, sometimes referred to as her type, is denoted by $v \in [\underline{v}, \overline{v}]$, with population distribution $\phi \in \Delta([\underline{v}, \overline{v}])$[5], assumed to have a strictly increasing and continously differentiable cumulative distribution function (CDF), F, on the interior of $[\underline{v}, \overline{v}]$. I normalize the median of F to be $v^m = 0$. Moreover, I assume voters' preferences to be symmetric around their ideal points.[6]

There are two parties, the Democratic and the Republican party. Party affiliation is determined by a threshold $k \in [\underline{v}, \overline{v}]$. For simplicity, I assume $[\underline{v}, \overline{v}]$ is big enough, that is $\underline{v} < -k$. In particular, I call Republicans the voters such that $v \geqslant k$ and Democrats the voters such that $v < k$.

Gerrymandering. A Republican designer, or redistricter, is in charge of creating equipopulous districts so as to maximize his party's seat share. She allocates voters among a continuum of districts based on their type v, thus determining the distribution $\pi \in \Delta([\underline{v}, \overline{v}])$, with CDF P, of voter types within a district. A redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$[7] is, therefore, a distribution over distributions: it specifies the measure of districts with each distribution π of voter types. To satisfy the equipopulous requirement typical of gerrymandering, any plan must be such that the following budget constraint holds:[8]

$$\int \pi d\mathcal{H}(\pi) = \phi. \tag{BC}$$

For instance, uniform redistricting imposes all districts to be the same, thus $\mathcal{H}(\phi) = 1$, while perfect segregation imposes that each voter type $v \in [\underline{v}, \overline{v}]$ constitutes a district on their own.

[5]For any complete and separable metric space X, I let $\Delta(X)$ denote the set of Borel probability measures on X, endowed with the weak* topology.

[6]For instance, a voter type v's preference for policy $x \in [\underline{v}, \overline{v}]$ could be represented by utility function $u(x; v) = -|x - v|$.

[7]Note that $\Delta([\underline{v}, \overline{v}]])$ is metrizable as a complete and separable metric space with the weak* topology.

[8]The integral sign is used to denote the Lebesgue integral as defined in Aliprantis and Border (1994).

Vote Shares. In any district with distribution $\pi \in \text{supp}(\mathcal{H})$,[9] two-stage elections are held. In primary elections, Republican voters choose a Republican candidate (R), while Democratic voters choose a Democratic candidate (D). During general elections, each voter chooses the candidate whose policy position is closer to their ideal point. The designer wins a district if and only if the Republican candidate receives at least a majority of the district vote.

I model electoral incentives so that D and R locate at the lowest median of $\pi(\cdot|v < k)$[10](the median of Democratic affiliates) and at the highest median of $\pi(\cdot|v \geqslant k)$ (the median of Republican affiliates), respectively. There are various reasons why primary candidates might cater to their respective party medians rather than converge to the overall district median. In the Supplemental Appendix, I provide a detailed model based on Owen and Grofman (2006), where both voters and candidates are uncertain about the position of the population median. However, it is important to note that the strict dependence on party medians is not necessary for my results to go through. For instance, the results are robust to any alternative rule that determines the position of candidate D (respectively, R) through a convex combination of the median of Democratic (respectively, Republican) affiliates and the population median. To close the model, I assume that whenever $\text{supp}(\pi) \subseteq [k, \infty)$ (respectively, $\text{supp}(\pi) \subseteq (-\infty, k]$), D (respectively, R) takes position k. This assumption intuitively captures the idea that, even in a district with a very high Democratic (respectively, Republican) majority, there is always an arbitrarily small fraction of the most moderate Republicans (respectively, Democrats) close to k. The latter assumption and the tie-breaking rules are discussed in the Supplemental Appendix C.

Formally, call $c_{\pi,D}$ and $c_{\pi,R}$ the position taken by D and R in district π. Then:

$$(c_{\pi,D}, c_{\pi,R}) = \begin{cases} (v^m_{\pi,D}, k) & if \ \text{supp}(\pi) \subseteq (-\infty, k] \\ (k, v^m_{\pi,R}) & if \ \text{supp}(\pi) \subseteq [k, \infty) \\ (v^m_{\pi,D}, v^m_{\pi,R}) & otherwise \end{cases},$$

[9]Throughout, $\text{supp}(\cdot)$ denotes the support of a probability measure, defined as the set of all points whose every open neighborhood has positive measure.

[10]Throughout, $\mu(Y|X)$ denotes the conditional probability distribution of Y given X according to measure μ.

where:

$$v^m_{\pi,D} = \inf_a \left\{a: \ \pi([a,\overline{v}]|v<k) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]|v<k) \geqslant \frac{1}{2}\right\}$$
$$v^m_{\pi,R} = \sup_a \left\{a: \ \pi([a,\overline{v}]|v\geqslant k) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]|v\geqslant k) \geqslant \frac{1}{2}\right\}.$$

Voters choose their district representative by majority rule. Hence, the candidate who wins the elections is the one closer to the district's median. Formally, the position of the district representative, c_π, is:

$$c_\pi = \begin{cases} c_{\pi,D} & if \ \ v^m_\pi < \frac{c_{\pi,R}+c_{\pi,D}}{2} \\ c_{\pi,R} & if \ \ v^m_\pi \geqslant \frac{c_{\pi,R}+c_{\pi,D}}{2} \end{cases}$$

where:

$$v^m_\pi = \sup_a \left\{a: \ \pi([a,\overline{v}]) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]) \geqslant \frac{1}{2}\right\}.$$

The designer wins district π if the winning candidate is the Republican candidate; that is, if $c_\pi \geqslant k$. Note that, in this model, the designer can win a district even without any Republican voters. This feature should be interpreted as a limit case that allows me to isolate the mechanism of interest while maintaining tractability. In Section 5, I show how different assumptions can be relaxed so that the designer organically needs a non-vanishing fraction of Republicans in each district to win.

Given redistricting plan $\mathcal{H}$, the designer's seat share is:

$$\int \mathbb{1}\left(c_\pi \geqslant k\right) d\mathcal{H}(\pi) = \int \mathbb{1}\left(v^m_\pi - \frac{c_{\pi,R}+c_{\pi,D}}{2} \geqslant 0\right) d\mathcal{H}(\pi).$$

Aggregate Uncertainty. Suppose that, after the designer commits to a plan, but before candidates choose their positions, an aggregate location shock affects all voters[11]. Formally, each voter experiences a common shock $\omega \in \mathbb{R}$, so that her ideal point becomes $v-\omega$. Assume that ω has distribution $\gamma \in \Delta(\mathbb{R})$, with CDF G, assumed to be Lipschitz continuous[12] and strictly increasing on $[\underline{v}-\overline{v}, 2\overline{v}-2\underline{v}]$, so that any district has a positive probability of being won or lost. Any shock ω induces a new preference distribution in each district. I call π^ω such induced distribution, with CDF $P^\omega = P(v+\omega)$.

[11]In Section 3.2, I argue that the case of a small shock is the empirically relevant one. Nevertheless, allowing for uncertainty serves the technical purpose of smoothing out the objective function and selecting a unique, robust solution from the multiplicity that exists without a shock.

[12]For instance, if G is everywhere continuously differentiable, it satisfies the assumptions.

Redistricter's Problem. The redistricter, or designer, wishes to maximize the expected seat share won by Republicans. Her problem is:

$$\max_{\mathcal{H}\in\Delta(\Delta([\underline{v},\overline{v}]))} \int\int \mathbb{1}\left(v^m_{\pi^\omega} - \frac{c_{\pi^\omega,D}+c_{\pi^\omega,R}}{2} \geqslant 0\right) d\mathcal{H}(\pi)d\gamma(\omega) \tag{RP}$$

$$s.t. \int \pi d\mathcal{H}(\pi) = \phi.$$

A redistricting plan $\mathcal{H}$ is *feasible* if it satisfies constraint (BC). It is *optimal* if it is a solution to (RP).

It is instructive to compare (RP) to a redistricting problem with exogenous policies, whose solution can be found with off-the-shelf tools from recent literature. Suppose that candidates' positions are fixed: $c_{\pi^\omega,R} = \overline{v}$, $c_{\pi^\omega,D} = \underline{v}$. The redistricter's problem under exogenous policies is:

$$\max_{\mathcal{H}\in\Delta(\Delta([\underline{v},\overline{v}]))} \int\int \mathbb{1}\left(v^m_{\pi^\omega} \geqslant k^\star\right) d\mathcal{H}(\pi)d\gamma(\omega) \tag{RPEx}$$

$$s.t. \int P d\mathcal{H}(\pi) = F.$$

where $k^\star = \frac{\underline{v}+\overline{v}}{2}$. The object of interest to the designer is an element of $\Delta(\Delta([\underline{v},\overline{v}]))$, a distribution over distributions. However, in the case of exogenous policies, the objective function only depends on district medians. Hence, the problem can be mapped to a maximization problem over elements of $\Delta([\underline{v},\overline{v}])$, which are much simpler objects to work with. The set of feasible distributions of medians can be characterized using Theorem 2 in Yang and Zentefis (2024). The transformed problem is:

$$\max_{\chi\in\Delta([\underline{v},\overline{v}])} \int\int \mathbb{1}\left(v-\omega \geqslant k^\star\right) d\chi(v)d\gamma(\omega)$$

$$s.t.\quad \max\{2F(v)-1,0\} \leqslant X(v) \leqslant \min\{2F(v),1\}, \text{ for all } v\in[\underline{v},\overline{v}],$$

where X is the CDF associated with $\chi\in\Delta([\underline{v},\overline{v}])$. Switching the order of integration, it can be rewritten as:

$$\max_{\chi\in\Delta([\underline{v},\overline{v}])} \int G(v-k^\star)d\chi(v)$$

$$s.t.\quad \max\{2F(v)-1,0\} \leqslant X(v) \leqslant \min\{2F(v),1\}, \text{ for all } v\in[\underline{v},\overline{v}].$$

Since the problem is linear in χ, it suffices to focus attention on the extreme points of the set of feasible distributions of medians. Again, Yang and Zentefis (2024) characterize such a set in Theorem 1. In my case, since G is increasing, the solution is $X^\star = \max\{2F-1, 0\}$, which dominates all other feasible distributions of medians, in the first-order-stochastic-dominance sense. We can now recover all solutions to (RPEx) as all those redistricting plans that induce $X^\star$. It is easy to see that there are many such plans. The following proposition, already proven with a different argument by Kolotilin and Wolitzky (2026), describes such plans and summarizes the result for exogenous policies.

Proposition 0. *A feasible redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ is a solution to the redistricting problem with exogenous policies (RPEx) if and only if, for all $\pi \in \operatorname{supp}(\mathcal{H})$, except for at most a zero-measure subset, there exists $v_\pi \geqslant 0$ such that $\pi(\{v_\pi\}) = \pi(\{v : v \leqslant 0\}) = \frac{1}{2}$.*

In the case of endogenous policies, it is not possible to reduce the object of interest to a distribution over a uni-dimensional space, because the objective function depends both on district medians and on candidates' positions. Nevertheless, in the next subsection, I show that the problem can still be simplified and solved using a different set of tools. In particular, I show that the object of interest can be reduced to a joint distribution over two uni-dimensional spaces, thus invoking the literature on the so-called optimal transport.

2.2 The Designer's Problem as an Optimal Transport Problem

While the redistricter's problem (RP) appears challenging, it can be simplified and restated as an optimal transport problem. The following result provides necessary conditions for optimality and constitutes the main ingredient for this transformation.

Proposition 1. *A feasible redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ is optimal only if, for all $\pi \in \operatorname{supp}(\mathcal{H})$, there exist $v' \geqslant 0$ and $v'' \leqslant 0$ such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$.*

Proposition 1 constrains the set of feasible plans: optimality requires that the population of any district has at most two types. Moreover, each district in an optimal plan must place half the mass on a voter type above the median of F and the rest on a voter type below the median of F. Effectively, this proposition states that (RP) is a matching problem of voter types across the median of their population distribution.

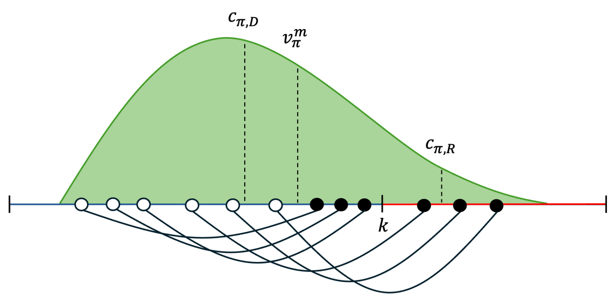
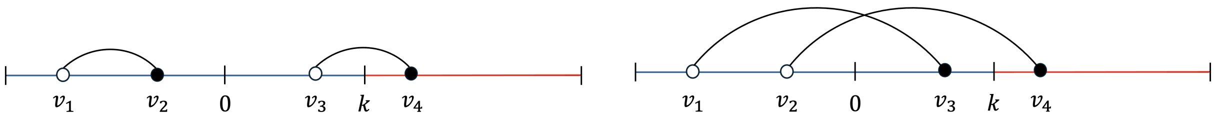


Figure 4: district π (in green) can be replaced by a continuum of binary districts. Each such district matches types above the median of π (●) to types below it (○) in a positive assortative manner.

The proof proceeds in two steps, for which I now provide an intuition.

First, it can be shown that any district in an optimal plan must be a *binary district*. That is, it must contain at most two voter types (in equal proportions). Consider a plan $\mathcal{H}$ and any district π in its support. As it turns out, $\mathcal{H}$ can be improved in the following way. Imagine replacing π with multiple smaller districts (possibly infinitely many), where each new district contains at most two voter types in equal proportions: a high type from above the median v_π^m and a low type from below v_π^m. Further, imagine that types above v_π^m and types below v_π^m are allocated to districts in a positive assortative manner. Specifically, suppose that any two districts contain types (h, l) and (h', l'), respectively. If the first district has a higher high type, $h > h'$, then it must also have a higher low type, $l \geqslant l'$. Such construction is illustrated in Figure 4. For illustration purposes, suppose that the shock realization is $\omega = 0$. Then, district π is lost, since v_π^m is closer to the Democratic candidate, $c_{\pi,D}$, than it is to the Republican candidate, $c_{\pi,R}$. Nevertheless, each of the binary districts depicted in Figure 4 is won. Indeed, consider any of the districts $\hat{\pi}$ where the highest type (●) is above k. Then, the Republican candidate $c_{\hat{\pi},R}$ sits exactly at the median $v_{\hat{\pi}}^m$, so that $c_{\hat{\pi},R} = v_{\hat{\pi}}^m =$ ●, and the district is won. Now consider any of the districts $\tilde{\pi}$ where the highest type is below k, so that $\tilde{\pi}$ contains only Democrats. Then, the Republican candidate $c_{\tilde{\pi},R}$ sits at k, the Democratic candidate $c_{\tilde{\pi},D}$ sits at the lowest type (○), and the highest type (●) constitutes the median voter $v_{\tilde{\pi}}^m$. Note that $c_{\tilde{\pi},R}$ is more moderate than $c_{\pi,R}$, $c_{\tilde{\pi},D}$ is more extreme than $c_{\pi,D}$, and $v_{\tilde{\pi}}^m$ is higher than v_π^m, so that each $\tilde{\pi}$ is won. The fact that high types (●) are matched to low types (○) in a positive assortative manner ensures that the above reasoning works no matter the realization of the shock ω.



(a) Two districts containing types from the same side of the population median.

(b) Alternative matching.

Figure 5: Districts must match voter types across the population median.

Second, binary districts must contain exactly two voter types: the highest type drawn from above 0, the population median, and the lowest type drawn from below 0. Figure 5 illustrates this point. Suppose there are two districts, one containing types v_4 and v_3 and the other containing types v_2 and v_1, as shown in Figure 5a. Consider the alternative matching shown in Figure 5b, where v_1 is paired with v_3, while v_2 is paired with v_4. It can be shown that the pairing in 5b is superior to the one in 5a. For illustration purposes, suppose the realization of the shock is $\omega = 0$. Consider the district containing v_4. In both configurations, the Republican candidate sits exactly at v_4 and receives exactly half the votes, enough to win the district. Now consider the district containing v_1. In both cases, the Democratic candidate sits at v_1 and the Republican candidate sits at k. In 5a, v_2 is closer to v_1 than to k, so that all Democrats vote for the Democratic candidate and the district is lost. However, in 5b, type v_3 prefers voting for the Republican candidate, so that the district is won with exactly half the votes.

Proposition 1 justifies the definition of a relation between the set of redistricting plans and the set of joint distributions over $[\underline{v}, 0] \times [0, \overline{v}]$. Define $\phi' = \phi(\cdot | v \geqslant 0)$ and $\phi'' = \phi(\cdot | v < 0)$. In words, ϕ' is the distribution of voter types conditional on them being above the population median, while ϕ'' is the distribution of voter types conditional on them being below the population median. I call $T(\phi', \phi'') \subseteq \Delta([\underline{v}, 0] \times [0, \overline{v}])$ the set of joint distributions having marginals ϕ' and ϕ''.[13] Following the literature on optimal transport, I sometimes refer to $T(\phi', \phi'')$ as the set of *transport plans* from ϕ' to ϕ''.

Call $\Delta_2 \subseteq \Delta(\Delta([\underline{v}, \overline{v}]))$ the set of feasible plans $\mathcal{H}$ such that, for all $\pi \in \text{supp}(\mathcal{H})$, there exist $v' \geqslant 0$ and $v'' \leqslant 0$, with $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$. There is a bijection from

[13]Formally, $T(\phi', \phi'') = \left\{\tau \in \Delta([\underline{v}, 0] \times [0, \overline{v}]) : \text{proj}_{[\underline{v}, 0]} \# \tau = \phi'', \text{proj}_{[0, \overline{v}]} \# \tau = \phi'\right\}$, where: $\text{proj}_{[\underline{v}, 0]}$ and $\text{proj}_{[0, \overline{v}]}$ denote the projection functions of $[\underline{v}, 0] \times [0, \overline{v}]$ on $[\underline{v}, 0]$ and $[0, \overline{v}]$, respectively; $(\text{proj}_{[\underline{v}, 0]} \# \tau)(A) = \tau(\text{proj}_{[\underline{v}, 0]}^{-1}(A))$ and $(\text{proj}_{[0, \overline{v}]} \# \tau)(A) = \tau(\text{proj}_{[0, \overline{v}]}^{-1}(A))$ for all A measurable.

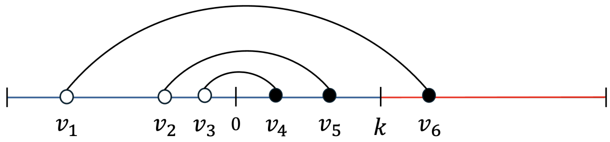


Figure 6: Of the three depicted districts, only the one containing voter types v_3 and v_4 is lost when $\omega = 0$.

$T(\phi', \phi'')$ to Δ_2. Moreover, note that any district placing positive mass on $v' \geqslant 0$ and $v'' \leqslant 0$ can be won in one of two ways. For instance, after the realization of the location shock ω, it can be that $v' - \omega$ is above k, so that the district has at least a majority of Republican affiliates. For such a district, I have $v^m_{\pi^\omega} = c_{\pi^\omega,R} = v' - \omega$ and $c_{\pi^\omega,D} = v'' - \omega$. Alternatively, it can be that $v' - \omega$ is less than k, but still closer to k than it is to $v'' - \omega$, so that the district is split fifty-fifty between "extreme" and "moderate" Democrats, the latter choosing the default moderate Republican candidate. For such a district, I have $v^m_{\pi^\omega} = v' - \omega$, $c_{\pi^\omega,R} = k$, and $c_{\pi^\omega,D} = v'' - \omega$, with $k - (v' - \omega) \leqslant (v' - \omega) - (v'' - \omega)$. In Figure 6, three types of districts are shown. When $\omega = 0$, only the district containing voter types v_3 and v_4 is lost.

Given the above discussion, I can rewrite (RP) as:

$$\max_{\tau \in T(\phi', \phi'')} \int\int \mathbb{1}(v' - \omega \geqslant k) + \mathbb{1}(v' - \omega < k)\mathbb{1}\left(v' - \omega - \frac{v'' - \omega}{2} \geqslant \frac{k}{2}\right) d\tau(v', v'') d\gamma(\omega).$$

By switching the order of integration and further manipulating the above, I get the optimal transport problem:

$$\max_{\tau \in T(\phi', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v''). \tag{OTP}$$

A transport plan τ in $T(\phi', \phi'')$ is *pure* whenever $(v', v'') \in \operatorname{supp}(\tau)$ implies $(v', \tilde{v}''), (\tilde{v}', v'') \notin \operatorname{supp}(\tau)$ for $v' \neq \tilde{v}'$ and $v'' \neq \tilde{v}''$. Intuitively, purity requires that no "splitting of masses" occurs across voter types. A pure plan is sometimes referred to as a *transport map*.

Define $T^\star \subseteq T(\phi', \phi'')$ as the set of solutions to (OTP) and $\Delta_2^\star \subseteq \Delta_2$ as the set of solutions to (RP) in Δ_2. The optimal transport problem (OTP) is *equivalent* to the redistricter's problem (RP) if there exists a bijection map from $T^\star$ to $\Delta_2^\star$. The following theorem summarizes the discussion in this subsection.

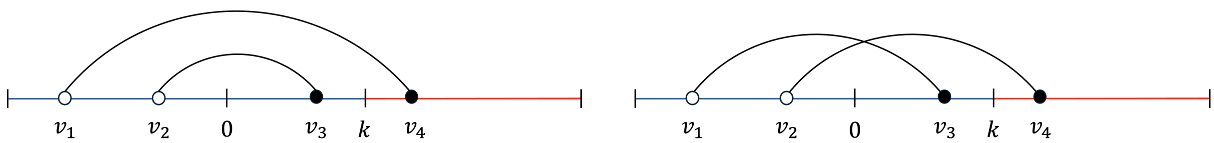
Theorem 1. *The optimal transport problem (OTP) is equivalent to the redistricter's problem (RP).*

3 Characterizing the Optimal Redistricting Plan

In this section, I characterize the solution(s) to the optimal transport problem (OTP) under different assumptions on the shock distribution G. The optimal redistricting plan depends heavily on the shape of G and, in most cases, on the distribution of voters F.

3.1 Benchmark Cases

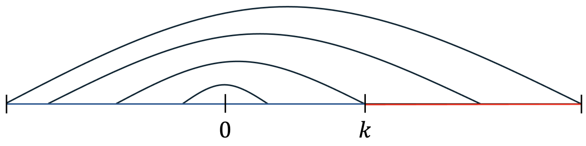
I start by describing a few benchmark cases before focusing on the most realistic case of a symmetric, S-shaped shock. Suppose that G is strictly convex on its support. The location shock "shifts" the distribution of preferences. A negative shock shifts it to the right, increasing the fraction of Republican voters (hence it is a favorable shock), while a positive shock shifts the distribution to the left (so it is unfavorable). When G is convex, the marginal benefit of slightly improving a competitive district's safety is outweighed by the cost of marginally reducing the safety of an already secure district. Consider Figure 7. Panel 7a depicts two districts, the one containing voter types v_1, v_4 considerably safer than the one containing types v_2, v_3. One can think of constructing two alternative districts of intermediate safety, as in panel 7b, by matching v_1 to v_3 and v_2 to v_4. Because G is convex, the designer always prefers the matching in 7*a* to the one in 7*b*. In other words, the designer wants to create very safe districts consisting of extremists of both parties, alongside very unsafe districts consisting of more moderate voters. Based on this intuition, it can be shown that there exists a unique, pure solution to OTP and that such solution maps types above zero to types below zero in a *negative*



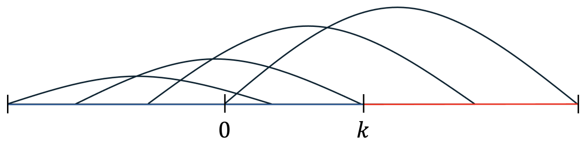
(a) The district containing voter types v_1, v_4 is considerably safer than the one containing types v_2, v_3.

(b) The two districts are of similar safety.

Figure 7: Two alternative configurations of districts.



(a) When G is convex, the optimal transport plan maps types above 0 to types below 0 in a negative assortative manner.



(b) When G is concave, the optimal transport plan maps types above 0 to types below 0 in a positive assortative manner.

Figure 8: The unique optimal plan under G convex/concave.

assortative manner[14]. Figure 8a illustrates such negative assortative map.

Concave Shock. The case of a strictly concave G is analogous to that of a strictly convex G. In this case, the marginal benefit of slightly improving a competitive district's safety exceeds the cost of marginally reducing the safety of an already secure district. Hence, the designer prefers the configuration in Figure 7b to the one in Figure 7a and tries to create districts of similar safety. This preference for more balanced districts creates a natural tendency against negative assortative matching. Indeed, it can be shown that there exists a unique, pure solution to OTP that maps types above 0 to types below 0 in a *positive assortative manner*,[15] as illustrated in Figure 8b.

Uniform Shock. For the sake of completeness, I analyze the case when G is affine on its support. It is easy to see that, given the linearity of G, problem OTP does not depend on T, therefore $T^\star = T(\phi', \phi'')$. In other words, any redistricting plan in Δ_2 is optimal.

The following result formalizes the findings for benchmark cases.

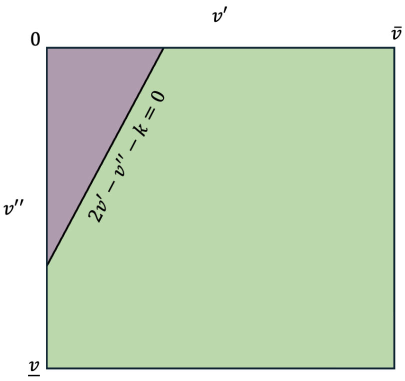
Proposition 2. *Consider the following cases:*

- *If G is strictly convex on its support, there exists a unique, pure solution to OTP and it is a negative assortative map.*
- *If G is strictly concave on its support, there exists a unique, pure solution to OTP and it is a positive assortative map.*
- *If G is affine on its support, any $\tau \in T(\phi', \phi'')$ is a solution to (OTP).*

[14] Formally, a *negative assortative map* $\tau \in T(\phi', \phi'')$ is such that, for all $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, $v' > \tilde{v}'$ implies $v'' \leqslant \tilde{v}''$.

[15] Formally, a *positive assortative map* $\tau \in T(\phi', \phi'')$ is such that, for all $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, $v' > \tilde{v}'$ implies $v'' \geqslant \tilde{v}''$.

To prove Proposition 2, I use results in the optimal transport literature as off-the-shelf tools. Such tools fail to be useful in cases where G is neither convex nor concave. In the next subsections, I show how the above benchmark results can still be used as building blocks to characterize the solutions to (OTP) when the shock is symmetric, strictly convex below zero, and strictly concave above zero, or S-shaped.



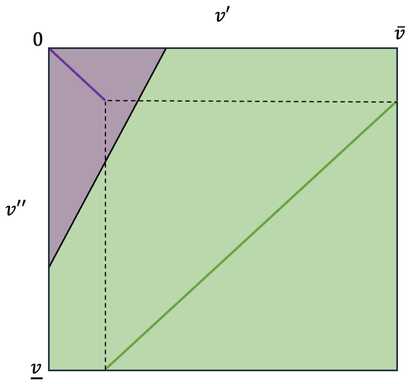


Figure 9: The support of $\tau \in T(\phi'.\phi'')$ is split into two regions by the line $2v'-v''-k=0$. Voters in the purple region are matched in a negative assortative manner, while voters in the green region are matching in a positve assortative manner.

3.2 S-shaped and Symmetric Shock

Suppose the shock is strictly S-shaped and symmetric around zero.[16] Any normal shock with mean zero falls under this category. As it turns out, (OTP) admits a unique solution. Building on the benchmark cases studied in the previous subsection, I show that such a solution is the convex combination of a positive assortative map and a negative assortative map, each over an appropriate subset of $[0,\overline{v}] \times [\underline{v},0]$. Figure 9 depicts $[0,\overline{v}] \times [\underline{v},0]$ and partitions it into two subsets. For any (v',v'') belonging to the purple region, $2v' - v'' - k$ is less than zero, while for any (v',v'') belonging to the green region, $2v' - v'' - k$ is greater than zero. Hence, G is convex on the purple subset of $[0,\overline{v}] \times [\underline{v},0]$ and concave on the green one. I show that any two couples of voter types falling in the green subset of $[0,\overline{v}] \times [\underline{v},0]$ must constitute a positive assortment, in order to be part of a solution to (OTP). Similarly, any two couples of voter types falling in the purple subset of $[0,\overline{v}] \times [\underline{v},0]$ must constitute a negative assortment. The following result formalizes such intuition.

[16]Formally, G is strictly convex on $[\underline{v},0]$ and strictly concave on $[0,\overline{v}]$. For all x, $G(x) = 1 - G(-x)$.

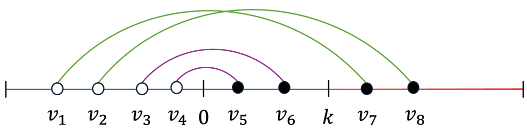
Proposition 3. *Suppose the aggregate shock is S-shaped and symmetric around* 0*. There exist* τ^+*,* $\tau^- \in \Delta([0,\overline{v}] \times [\underline{v},0])$ *and* $\alpha \in [0,1]$ *such that:*

1. τ^- *is negative assortative with* $\operatorname{supp}(\tau^-) \subseteq \{(v',v'') \in [0,\overline{v}] \times [\underline{v},0] : 2v' - v'' - k \leqslant 0\}$

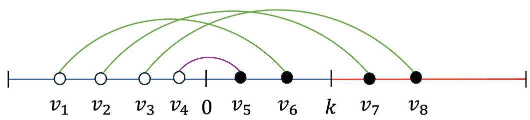
2. τ^+ *is positive assortative with* $\operatorname{supp}(\tau^+) \subseteq \{(v',v'') \in [0,\overline{v}] \times [\underline{v},0] : 2v' - v'' - k \geqslant 0\}$

3. $(1-\alpha)\tau^- + \alpha\tau^+$ *is the unique solution to OTP.*

Call $T^{\pm}$ the set of plans $\tau = \alpha\tau^+ + (1-\alpha)\tau^-$, where τ^+ and τ^- are as in Proposition 3. Further characteristics of the solution depend on the precise shapes of F, the distribution of voters, and of G, the distribution of the shock. A particularly important case is that of a small aggregate shock, as supported by recent evidence based on precinct-level returns from the 2016, 2018, and 2020 U.S. House elections (Kolotilin and Wolitzky, 2026). What happens in that case? Consider Figure 10. Two plans, both belonging to $T^{\pm}$, are depicted. In Figure 10a, voter types v_1, v_2, v_7, and v_8 are matched positively assortatively, while v_3, v_4, v_5, and v_6 are matched negatively assortatively; in Figure 10b, voter types v_1, v_2, v_3, v_6, v_7 and v_8 are matched positively assortatively, while only v_4 and v_5 are matched negatively assortatively. The latter configuration is obtained from the former by breaking the match between v_3 and v_6 and including them in the positive assortative match. Which plan is preferred by the designer? On one hand, Figure 10b introduces an additional match where $2v' - v'' - k > 0$. On the other hand, it reduces the magnitude by which this inequality is satisfied by the other matches in green. If the variance of the shock is small enough, the configuration in Figure 10b proves superior to the one in Figure 10a. This follows from the fact that as the variance of the shock decreases, G becomes steeper around zero than at any positive value. Then, the marginal benefit of converting a negative assortative match into



(a) Two positive assortative matched (in green) and two negative ones (in purple).



(b) Three positive assortative matched (in green) and one negative (in purple).

Figure 10: If the variance of the shock is small enough, the configuration in the bottom panel is preferred to the one in the top panel.

a positive assortative one is larger than the marginal cost of reducing the degree of positivity in existing green matches (as long as they stay positive). Call σ the variance of the shock with CDF G_σ, with the usual assumptions, and $\tau^\star_\sigma$ the solution to (OTP) when the shock has variance σ. Let $\alpha_{max} = \max_{\tau \in T(\phi', \phi'')} \tau(\{(v', v'') : 2v' - v'' - k \geqslant 0\})$. The following holds.[17]

Proposition 4. *Suppose the shock is symmetric and S-shaped around* 0*. As* $\sigma \to 0$*,* $\tau^\star_\sigma \to \tau^\star$*, where* $\tau^\star(\{(v', v'') : 2v' - v'' - k \geqslant 0\}) = \alpha_{max}$*.*

Call $\alpha^\star_\sigma$ the mass of positive assortative matches under $\tau^\star_\sigma$. An immediate corollary follows.

Corollary 1. *For all* $\epsilon > 0$*,*

$$\liminf_{\sigma \to 0} \alpha^\star_\sigma \geqslant \alpha_{max} - \epsilon.$$

Moreover, if $\tau^\star(\{(v', v'') : 2v' - v'' - k = 0\}) = 0$*,* $\lim_{\sigma \to 0} \alpha^\star_\sigma = \alpha_{max}$*.*

In words, as the variance of the shock becomes smaller and smaller, the solution favors positive assortative matches more and more.

Finally, note that the optimal plan is likely not pure, meaning that some voter types might belong to both the green and the purple region simultaneously. Figure 11 shows simulated solutions for fixed G and F, both normal with variance 1, for different values of k. Regions of non-purity are evident. For instance, in the second panel ($k = 2.5$), some voters with type $v'' = -0.5$ are matched to voters of type $v' < 1$, while others are matched to voters of type $v' > 2$. Not surprisingly, as k increases and the fraction of Republicans in the population decreases, the designer is worse off. Indeed, the line $2v' - v'' - k = 0$ shifts down and the region of positive assortative matches shrinks. In the Supplemental Appendix E, I provide additional simulations for varying values of σ and k.

[17] The proof exploits convergence of the hypograph of $G(x)$ to that of $\mathbb{1}(x \geqslant 0)$ and a variational argument.

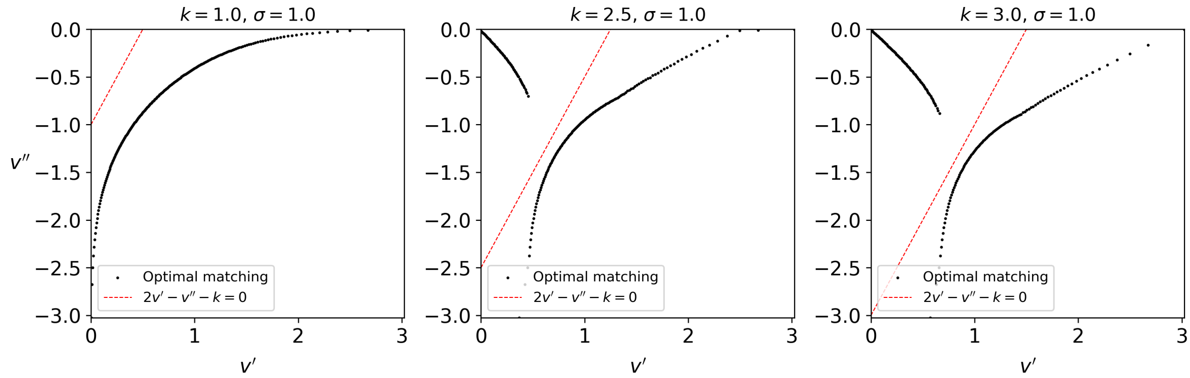


Figure 11: Simulated solution for F and G normal with variance 1, for different values of k.

4 Political and Legal Implications

Because of its effects on democratic representation and political polarization, gerrymandering has been controversial for decades. In this section, I discuss three implications of my model for current political and legal debates.

4.1 Electorate Polarization and Effectiveness of Gerrymandering

One can ask what happens when the "spread" of the voter distribution changes. If the tails become fatter, so that more extreme voters appear on both sides of the spectrum, does the designer benefit? More precisely, consider a new distribution of voters $\tilde{\phi}$, a median-preserving spread (Definition 5 in the Appendix) of the original ϕ, and subject to the same assumptions. The following holds.

Proposition 5. *Suppose $\tilde{\phi}$ is a median-preserving spread of ϕ. The expected seat-share won by the designer under $\tilde{\phi}$ is larger than under ϕ.*

Importantly, the above is true even if the fraction of Republicans in the population stays the same, that is $F(k) = \tilde{F}(k)$. The intuition is that spreading mass from the center to the tails increases the ideological distance between voter types at the population level. In particular, it widens the gap between moderate and extreme Democrats, giving the designer more room to exploit this wedge through redistricting.

4.2 Gerrymandering and Congress Polarization

The relationship between gerrymandering and partisan polarization in the U.S. House of Representatives is more complex than public discourse suggests. Gerrymandering is often blamed for heightened partisanship and legislative gridlock, but empirical research remains divided. McCarty et al. (2009) question its significance, while others find measurable effects (Kenny et al., 2023). Theoretical models with exogenous candidates yield conflicting predictions. Models with segregation of extreme opponents (Gul and Pesendorfer, 2010; Kolotilin and Wolitzky, 2026) predict a "gap" in the ideological distribution of elected representatives, with moderate supporters of the redistricting party at one end and extreme opponents at the other. Models without segregation of extreme opponents (Friedman and Holden, 2008) predict more district heterogeneity and a more continuous spectrum of representation. My model, which incorporates endogenous candidates emergence, predicts a gap in the distribution of district representatives without relying on segregation of extreme opponents. Let $\rho^{\omega}_{\mathcal{H}}$ denote the distribution of district representatives given a redistricting plan $\mathcal{H}$ and a shock realization ω. Formally:

Proposition 6. *For any optimal plan $\mathcal{H}$ and shock realization ω, $\rho^{\omega}_{\mathcal{H}}((-\omega, k)) = 0$, whenever $-\omega < k$.*

Like previous models, optimal plans create many right-leaning districts that elect moderate Republicans. However, opposition voters are never homogeneous in a district. An optimal plan creates districts with the right mix of moderate and extreme Democrats, with two consequences: first, extreme Democratic candidates win primaries by outnumbering moderates; second, moderate Democrats in these districts are effectively disenfranchised: they lack representatives who share their views and end up voting for the Republican candidate. As Proposition 1 shows, these disenfranchised moderate Democratic types fall in the interval $(-\omega, k)$, and never end up having their own representation. Importantly, this polarization is *asymmetric* across parties: the redistricting party elects moderate candidates, while the opposition party ends up represented by extremists. This asymmetry arises endogenously from the redistricter's optimal strategy, which deliberately induces extreme candidates among the opposition through primary elections. Unpublished evidence by Fowler et al. (2025) is consistent with this prediction, documenting asymmetric effects of partisan redistricting on the ideological positions of elected representatives.

4.3 Legislative implications for "majority-minority" districts

American federal legislation, including the 1965 Voting Rights Act, mandates that electoral district lines cannot be drawn to improperly dilute minorities' voting power. Since the 1986 Supreme Court decision in *Thornburg v. Gingles*, such laws have been interpreted as requiring the creation of districts where racial and ethnic minorities have the concrete opportunity to elect their own representatives. As a consequence, in the 118th Congress there are 26 congressional districts where Black people constitute a strict majority, and 37 are majority Hispanic or Latino (Klein, 2023).

While such legislation aims to increase minority representation, it has sparked debate over its broader political implications. The economic theory literature is divided: models predicting segregation of extreme opponents suggest majority-minority districts benefit Republicans, while models favoring heterogeneous districts imply these districts disrupt Republicans' desired optimum. In my model, majority-minority districts prevent the designer from exploiting the endogeneity of electoral incentives.

In some districts, minorities constitute a plurality rather than a majority. These "minority opportunity" districts provide minorities the chance to elect preferred representatives through coalitions with White voters. However, the success of this strategy depends on White Democrats' willingness to support candidates of color. If they are reluctant, a phenomenon known as "White backlash" may occur, potentially benefiting Republican candidates as described in this paper. A similar observation is made by Groll and O'Halloran (2025), who explicitly focus on maximizing minority benefits through redistricting.

5 Extensions

5.1 Quantile Redistricting

In the previous sections, I relied on party candidates whose positioning rule is a function of party medians. The implication is that the designer can win a district even if there are no supporters inhabiting it. I now consider a scenario where candidates' positions are determined by a quantile, not necessarily the median, of the distribution of preferences within each party. Let $\frac{1}{2} < q < 1$ be the quantile parameter. Under this setting, the Democratic candidate locates at the q-quantile of the preference distribution conditional on being affiliated with the Democratic party, while the Republican candidate locates at the $(1-q)$-quantile of the preference distribution conditional on being affiliated with the Republican party. In this setting, as it will become clear soon, the designer always needs a non-zero measure of supporters in order to win a district. Therefore, it becomes harder to characterize the solution in general[18]. For illustrative purposes, I assume F to be uniform on $[\underline{v}, \overline{v}]$, with $v^m = 0$ and $q \leqslant \frac{2}{3}$.

Call $c^q_{\pi,D}$ and $c^q_{\pi,R}$ the positions taken by D and R in district π. Then:

$$(c^q_{\pi,D}, c^q_{\pi,R}) = \begin{cases} (v^q_{\pi,D}, k) & if\ \operatorname{supp}(\pi) \subseteq (-\infty, k] \\ (k, v^q_{\pi,R}) & if\ \operatorname{supp}(\pi) \subseteq [k, \infty) \\ (v^q_{\pi,D}, v^q_{\pi,R}) & otherwise \end{cases}$$

where:

$$\begin{aligned} v^q_{\pi,D} &= \inf_a \quad \{a :\ \pi([a,\overline{v}] | v < k) \geqslant q\} \cap \{a :\ \pi([\underline{v}, a] | v < k) \geqslant q\} \\ v^q_{\pi,R} &= \sup_a \quad \{a :\ \pi([a,\overline{v}] | v \geqslant k) \geqslant 1-q\} \cap \{a :\ \pi([\underline{v}, a] | v \geqslant k) \geqslant 1-q\}. \end{aligned}$$

Every other detail of the model is the same as in Section 2. The following proposition characterizes the optimal measure of districts won by the designer for each realization of the shock ω, denoted by $V^\omega(q)$.

Proposition 7. *There are two cases:*

1. *Suppose $F(k+\omega) \leqslant \frac{1}{2q}$. Then $V^\omega(q) = 1$.*

2. *Suppose $F(k+\omega) > \frac{1}{2q}$. Then $V^\omega(q) = \frac{2q}{2q-1}(1 - F(k+\omega))$.*

In the first case, where the fraction of Democratic voters is less than or equal to $\frac{1}{2q}$, all districts are won by the Republican party. In the second case, where the fraction of Democratic voters is greater than $\frac{1}{2q}$, the designer wins $\frac{2q}{2q-1}(1-F(k+\omega))$ district, which is strictly more than $2(1-F(k+\omega))$, the districts she would win under exogenous

[18]For starters, the connection to optimal transport would probably require considering joint distributions with three fixed marginals, rather than just two. Hence, the optimal transport problem itself would become less tractable.

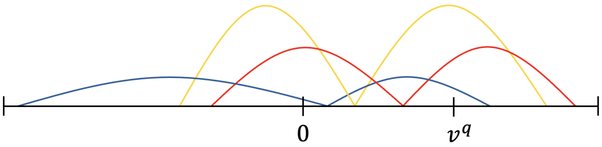


Figure 12: Three-wise positive assortative matching with probability masses of $\frac{1}{2}$, $\frac{1-q}{2q}$, and $\frac{2q-1}{2q}$.

policies. The intuition for the optimal plan is similar to the one in the previous sections. In each district, the redistricter needs to create a wedge between moderate and extreme Democrats, encouraging the emergence of an extreme Democratic candidate and counting on moderate Democrats to vote for a moderate Republican candidate. The following result characterizes optimal redistricting plans:

Proposition 8. *Define $v^q = F^{-1}\left(\frac{1}{2q}\right)$. A feasible redistricting plan $\mathcal{H}$ is optimal if and only if, for all $\pi \in supp\ (\mathcal{H})$, there exists $v'_\pi \geqslant v^q$ such that:*

1. $\pi(\{v'_\pi\}) = 1 - \pi(\{v : v < v^q\}) = \frac{2q-1}{2q}$, *and*

2. $|v^q_{\pi,D} - v^m_\pi| \geqslant |v^m_\pi - v'_\pi|$.

This proposition states that an optimal redistricting plan under quantile redistricting must satisfy two conditions for each district π in the support of the plan. First, the mass of voters with type exactly equal to v'_π should be $\frac{2q-1}{2q}$, and the mass of voters with type below v_q should be $1 - \frac{2q-1}{2q}$. Second, the distance between $v^q_{\pi,D}$, which is the lowest median of the district, and the highest median of the district should be greater than or equal to the distance between the highest median and v'_π. A redistricting plan that I call "three-wise positive assortative" plan complies with the requirements of Proposition 8. Figure 12 illustrates such a plan. Each district contains exactly three voter types, one type below 0, one type between 0 and v^q, and another type above v^q, with probability masses of $\frac{1}{2}$, $\frac{1-q}{2q}$, and $\frac{2q-1}{2q}$, respectively. Given shock ω, the designer wins district π if and only if $v'_\pi - \omega \geqslant k$.[19]

[19]While the results in this subsection use F uniform and $q \leqslant \frac{2}{3}$, this is not a necessary condition. The results hold for any distribution such that positive assortative matching of $v' \in [v^q, \overline{v}]$ to $v'' \in [0, v^q]$ and to $v''' \in [\underline{v}, 0]$ is such that $v' - v'' \leqslant v'' - v'''$.

5.2 Endogenous Turnout and other Alternative Mechanisms

This paper shows that gerrymandering can become more powerful when district composition affects voting behavior. While I have focused on the primary elections channel—where the designer exploits primaries to induce extreme candidates and convert moderate opposition voters—the core insight extends to other mechanisms, like endogenous turnout. For instance, suppose that whenever a Democratic voter is closer to the Republican candidate than to the Democratic one, instead of switching sides, they abstain from general elections. Let every other detail of the main model stay the same. As for the case of quantile redistricting, now the designer needs a positive fraction of Republicans (at least $\frac{1}{3}$) in the district to win it. The designer's objective is now to create districts where moderate Democrats become sufficiently discouraged by extreme Democratic candidates that they abstain rather than vote. The optimal strategy remains similar: pair extreme and moderate opposition voters to exploit the wedge between them, but keep enough Republicans in the district to win. For simplicity, suppose again that F is uniform on $[\underline{v}, \overline{v}]$. Consider the three-wise positive assortative plan that matches type v' in $[F^{-1}\left(\frac{2}{3}\right), \overline{v}]$ to type v'' in $[F^{-1}\left(\frac{1}{3}\right), F^{-1}\left(\frac{2}{3}\right)]$ to type v''' in $[\underline{v}, F^{-1}\left(\frac{1}{3}\right)]$, each with mass $\frac{1}{3}$, in a positive assortative manner. Because $v' - v'' \leqslant v'' - v'''$, each district is won whenever $v' \geqslant k + \omega$, meaning that for any realization of the shock ω, the designer wins $\min\{1, 3(1 - F(k + \omega))\}$ districts, which is more than he can win with full turnout and exogenous support. In this case, the designer is exploiting the endogeneity in turnout, rather than in partisanship, to his own advantage. Of course, one can model endogenous turnout in different ways, and at different stages of the election process (primary or general elections). One could also model endogenous district turnout based on pivotality considerations. Some such mechanisms lead to only marginally different models and solutions, while others merit deeper exploration in future work—as do other channels through which district composition shapes voter behavior, including information acquisition, campaign contributions, or grassroots mobilization.

6 Conclusions

This paper studies partisan gerrymandering when candidates' policy positions respond endogenously to district composition. The optimal strategy differs fundamentally from traditional pack-and-crack: rather than segregating extreme opponents or matching them with extreme supporters, the designer creates "mismatched slices" — pairing extreme with moderate opposition voters to exploit the ideological wedge between them. Primary election dynamics drive opposition candidates toward extreme positions, allowing the redistricting party to win districts by appealing to moderates. It follows that, when candidate positions respond to district composition, gerrymandering becomes more potent than previously understood.

The model yields several implications. First, if the electorate is more polarized around the median, gerrymandering becomes stronger. Second, the optimal plan produces an asymmetric ideological gap in Congress: the redistricting party elects moderate candidates, while the opposition is represented by extremists. This polarization emerges in heterogeneous districts, challenging the conventional view that political extremism stems primarily from voter segregation. Third, the model speaks to minority representation, particularly in ”minority opportunity” districts where success depends on coalition-building with white voters.

Several directions for future research remain, including incorporating idiosyncratic uncertainty about voter preferences, exploring alternative objective functions, and investigating other channels through which district composition shapes voter behavior. Across all such extensions, the fundamental insight persists: redistricting affects voting behavior, and designers can exploit it to their advantage.

A Proofs of Main Text

A.1 Proofs of Section 2

Proof of Proposition 0. Since the designer wins district $\pi \in \Delta([\underline{v},\overline{v}])$ whenever $v_\pi^m \geqslant k^\star + \omega$, a redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v},\overline{v}]))$ can be described by a distribution $\chi \in \Delta([\underline{v},\overline{v}])$ over $v = v_\pi^m$. Using Theorem 2 in Yang and Zentefis (2024), (RPEx) can be stated as:

$$\max_{\chi \in \Delta([\underline{v},\overline{v}])} \int\int \mathbb{1}\left(v - \omega \geqslant k^\star\right) d\chi(v) d\gamma(\omega)$$

$$s.t. \quad \max\{2F(v)-1,0\} \leqslant X(v) \leqslant \min\{2F(v),1\}, \text{ for all } v \in [\underline{v},\overline{v}].$$

Switching the order of integration, it can be rewritten as:

$$\max_{\chi \in \Delta([\underline{v},\overline{v}])} \int G(v-k^\star) d\chi(v)$$

$$s.t. \quad \max\{2F(v)-1,0\} \leqslant X(v) \leqslant \min\{2F(v),1\}, \text{ for all } v \in [\underline{v},\overline{v}].$$

By definition of first order stochastic dominance, and since G is strictly increasing, I have:

$$\int G(v-k^\star) d\chi(v) < \int G(v-k^\star) d\overline{\chi}$$

for any $\chi > \overline{\chi}$, where $\max\{2F(v)-1,0\}$ is the CDF associated with $\overline{\chi}$. Hence, the optimal distribution of medians is $X^\star = \max\{2F-1,0\}$. ■

Proof of Proposition 1. The proof relies on the following lemmas.

Lemma 1. *Suppose $\mathcal{H}$ is optimal. Then, for all $\pi \in \operatorname{supp}(\mathcal{H})$ (except for at most a zero-measure subset), either $|\operatorname{supp}(\pi)| = 1$, or there exist v', v'' such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$.*

Proof. The proof proceeds in two steps.

Step 1. I show that for any feasible plan $\mathcal{H}$, there exists a feasible plan $\hat{\mathcal{H}}$, such that for all $\hat{\pi} \in \operatorname{supp}(\hat{\mathcal{H}})$, either $|\operatorname{supp}(\hat{\pi})| = 1$, or there exist v', v'' such that $\hat{\pi}(\{v'\}) = \hat{\pi}(\{v''\}) = \frac{1}{2}$, and such that for all $\omega \in \operatorname{supp}(\gamma)$:

$$\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\hat{\pi}) \geqslant \int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\mathcal{H}(\pi).$$

Take any plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that $\int \pi d\mathcal{H}(\pi) = \phi$. First, for any $\pi \in \text{supp}(\mathcal{H})$, construct a measure $\hat{\mathcal{P}}_\pi \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that $\int \hat{\pi} d\hat{\mathcal{P}}_\pi(\hat{\pi}) = \pi$, and, for all $\hat{\pi} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $\text{supp}(\hat{\pi}) = \{v'_{\hat{\pi}}, v''_{\hat{\pi}}\}$, for some $v'_{\hat{\pi}}, v''_{\hat{\pi}} \in \text{supp}(\pi)$ with:

$$v'_{\hat{\pi}} \geqslant v^m_\pi,$$
$$v''_{\hat{\pi}} \leqslant v^m_\pi,$$
$$\hat{\pi}(\{v'_{\hat{\pi}}\}) = \hat{\pi}(\{v''_{\hat{\pi}}\}).$$

Moreover, let $\hat{\mathcal{P}}_\pi$ be positive assortative: for any $\hat{\pi}$, $\hat{\rho} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $v'_{\hat{\pi}} > v'_{\hat{\rho}} \implies v''_{\hat{\pi}} \geqslant v''_{\hat{\rho}}$. Define $P' = \max\{2P - 1, 0\}$ and $P'' = \min\{2P, 1\}$, where P is the CDF associated with π and note:

$$P'(x) = \int \mathbb{1}(v'_{\hat{\pi}} \leqslant x) d\hat{\mathcal{P}}_\pi(\hat{\pi}), \;\; P''(x) = \int \mathbb{1}(v''_{\hat{\pi}} \leqslant x) d\hat{\mathcal{P}}_\pi(\hat{\pi}).$$

Second, construct alternative plan $\hat{\mathcal{H}} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that, for any measurable set $A \subseteq \Delta([\underline{v}, \overline{v}])$, $\hat{\mathcal{H}}(A) = \int \hat{\mathcal{P}}_\pi(A) d\mathcal{H}(\pi)$. By construction, $\hat{\mathcal{H}}$ is feasible. I now show that:

$$\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\hat{\pi}) \geqslant \int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\mathcal{H}(\pi).$$

Specifically, I show that, for any $\pi \in \text{supp}(\mathcal{H})$, $\hat{\pi} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $\omega \in \text{supp}(\gamma)$, I have that $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

Consider the following cases:

1. If $v^m_{\pi^\omega} \geqslant k$, then $c_{\hat{\pi}^\omega, R} = v^m_{\hat{\pi}^\omega} \geqslant k$, which means that $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = 1 \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

2. If $v^m_{\pi^\omega} < k$ and $\text{supp}(\hat{\pi}^\omega) = \{v^m_{\pi^\omega}\}$, it must be that, for all $\hat{\rho} \in \text{supp}(\hat{\mathcal{P}}_\pi)$:

$$v''_{\hat{\rho}} < v''_{\hat{\pi}} = v^m_\pi \implies v'_{\hat{\rho}} \leqslant v'_{\hat{\pi}} = v^m_\pi,$$

   which means that $v''_{\hat{\rho}} < v^m_\pi \implies v'_{\hat{\rho}} = v^m_\pi$, where the equality holds because $v'_{\hat{\rho}} \geqslant v^m_\pi$. Then, it must be that $v^m_{\pi^\omega} = c_{\pi^\omega, D}$ and thus $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) = 0$.

3. If $v^m_{\pi^\omega} < k$, $|\text{supp}(\hat{\pi}^\omega)| = 2$, and $v^m_{\hat{\pi}^\omega} = c_{\hat{\pi}^\omega, R} \geqslant k$, then $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = 1 \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

4. If $v^m_{\pi^\omega} < k$, $|\text{supp}(\hat{\pi}^\omega)| = 2$, and $v^m_{\hat{\pi}^\omega} < k$, it must be that $c_{\hat{\pi}^\omega, R} = k \leqslant c_{\pi^\omega, R}$, and $v^m_{\hat{\pi}^\omega} \geqslant v^m_{\pi^\omega}$. Moreover, by definition of $c_{\pi, D}$ and the fact that $P = \frac{1}{2}P' + \frac{1}{2}P''$:

$$\pi''(\{v : c_{\pi^\omega, D} + \omega < v \leqslant v^m_\pi + \omega\}) \leqslant \frac{\pi'(\{v : v \geqslant k + \omega\})}{2},$$

where π', π'' are the measures associated with CDFs P', P''. By the positive assortative construction of $\hat{\mathcal{P}}_\pi$, it follows that $c_{\hat{\pi}^\omega,D} \leqslant c_{\pi^\omega,D}$ and thus $\mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k) \geqslant \mathbb{1}(c_{\pi^\omega} \geqslant k)$.

Step 2. I show that if $\mathcal{H}$ is optimal it must be that, for all $\pi \in \operatorname{supp}(\mathcal{H})$ (except at most for a zero-measure subset), either $|\operatorname{supp}(\pi)| = 1$, or there exist $v' \neq v''$ such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$.

Consider plan $\mathcal{H} \in \Delta(\Delta([\underline{v},\overline{v}]))$ such that $\int d\pi \mathcal{H}(\pi) = \phi$. Consider the construction in Step 1. For any $\pi \in \operatorname{supp}(\mathcal{H})$, $\hat{\pi} \in \operatorname{supp}(\hat{\mathcal{P}}_\pi)$, $\omega \in \operatorname{supp}(\gamma)$, I have that $\mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k) \geqslant \mathbb{1}(c_{\pi^\omega} \geqslant k)$. Suppose that, either $|\operatorname{supp}(\pi)| > 2$ or $\operatorname{supp}(\pi) = \{v', v''\}$ with $v' \neq v''$ and $\pi(\{v'\}) \neq \pi(\{v''\})$. I first show that:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega).$$

Call $\overline{v}_\pi$ the highest point in the $\operatorname{supp}(\pi)$.

Suppose $\overline{v}_\pi > v_\pi^m$. Take c such that $0 < c < \overline{v}_\pi - v_\pi^m$. Then, $\overline{v}_\pi - c > v_\pi^m$ and $\pi([\overline{v}_\pi - c, \overline{v}_\pi]) > 0$. Take $\omega^{\star\star}$ such that $k - \overline{v}_{\pi^{\omega^{\star\star}}} = \overline{v}_{\pi^{\omega^{\star\star}}} - c_{\pi^{\omega^{\star\star}},D}$. Take $d = \min\left\{c, \frac{k - \overline{v}_{\pi^{\omega^{\star\star}}}}{4}\right\}$. Take $\omega^\star = \omega^{\star\star} - 2d$. Then, all $\hat{\pi} \in \operatorname{supp}(\hat{\mathcal{P}}_\pi)$ with $v'_{\hat{\pi}} \in [\overline{v}_\pi - d, \overline{v}_\pi]$ are such that $\mathbb{1}(c_{\hat{\pi}^{\omega^\star}} \geqslant k) = 1$, while $\mathbb{1}(c_{\pi^{\omega^\star}} \geqslant k) = 0$. Hence, for all $\omega \in (\omega^\star, \omega^{\star\star})$, I have:

$$\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) > \mathbb{1}(c_{\pi^\omega} \geqslant k),$$

and thus:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega).$$

.

Suppose $\overline{v}_\pi = v_\pi^m$. Call $\underline{v}_\pi$ the lowest point in $\operatorname{supp}(\pi)$. Because $|\operatorname{supp}(\pi)| > 1$, $\underline{v}_\pi < v_\pi^m$. Take $\omega^{\star\star}$ such that $k - (\overline{v}_\pi - \omega^{\star\star}) = \overline{v}_\pi - \underline{v}_\pi$. Because either $|\operatorname{supp}(\pi)| > 2$ or $|\operatorname{supp}(\pi)| = 2$ with $\pi(\{v_\pi^m\}) > \frac{1}{2}$, I have that $\underline{v}_\pi - \omega^{\star\star} < c_{\pi^{\omega^{\star\star}},D}$. Take c such that $0 < c < c_{\pi^{\omega^{\star\star}},D} - (\underline{v}_\pi - \omega^{\star\star})$. Note that $\pi([\underline{v}_\pi - \omega^{\star\star}, \underline{v}_\pi - \omega^{\star\star} + c]) > 0$. Take $d = \min\left\{c, \frac{k - (\overline{v}_\pi - \omega^{\star\star})}{4}\right\}$. Take $\omega^\star = \omega^{\star\star} - d$. Then, all $\hat{\pi} \in \operatorname{supp}(\hat{\mathcal{P}}_\pi)$ with $v''_{\hat{\pi}} \in [\underline{v}_\pi, \underline{v}_\pi + d]$ are such that $\mathbb{1}(c_{\hat{\pi}^{\omega^\star}} \geqslant k) = 1$, but $\mathbb{1}(c_{\pi^{\omega^\star}} \geqslant k) = 0$. Hence, for all $\omega \in (\omega^\star, \omega^{\star\star})$, I have:

$$\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) > \mathbb{1}(c_{\pi^\omega} \geqslant k),$$

and thus:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega).$$

.

Now, suppose that there is a positive-measure subset of $\pi \in \mathrm{supp}(\mathcal{H})$ such that either $|\mathrm{supp}(\pi)| > 2$ or $\mathrm{supp}(\pi) = \{v', v''\}$ with $v' \neq v''$ and $\pi(\{v'\}) \neq \pi(\{v''\})$. I have:

$$\int\int\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) d\mathcal{H}(\pi) > \int\int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\gamma(\omega) d\mathcal{H}(\pi),$$

and:

$$\int\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\hat{\pi}) d\gamma(\omega) = \int\int\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\mathcal{H}(\pi) d\gamma(\omega) >$$
$$> \int\int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\gamma(\omega) d\mathcal{H}(\pi).$$

□

Lemma 2. *Suppose $\mathcal{H}$ is optimal and satisfies Lemma 1, that is, for all $\pi \in \mathrm{supp}(\mathcal{H})$ (except for at most a zero-measure subset), either $|\mathrm{supp}(\pi)| = 1$, or there exist v', v'' such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$. Then, for any such π (except for at most a zero measure subset), $v' \geqslant v^m$ and $v'' \leqslant v^m$.*

Proof. Suppose there exists $\overline{S} \subseteq \mathrm{supp}(\mathcal{H})$ such that $\mathcal{H}(\overline{S}) > 0$ and for all $\overline{\pi} \in \overline{S}$, $\mathrm{supp}(\overline{\pi}) = \{v'_{\overline{\pi}}, v''_{\overline{\pi}}\}$ with $v'_{\overline{\pi}} \geqslant v''_{\overline{\pi}} > v^m$. Since $\int \pi d\mathcal{H}(\pi) = \phi$, there must exist $\underline{S} \subseteq \mathrm{supp}(\mathcal{H})$ such that $\mathcal{H}(\underline{S}) = \mathcal{H}(\overline{S}) = p > 0$ and for all $\underline{\pi} \in \underline{S}$, $\mathrm{supp}(\underline{\pi}) = \{v'_{\underline{\pi}}, v''_{\underline{\pi}}\}$ with $v''_{\underline{\pi}} \leqslant v'_{\underline{\pi}} < v^m$.

Take any $\overline{\mathcal{H}} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ to be such that:

1. For any $\pi \in \mathrm{supp}(\overline{\mathcal{H}})$, $\mathrm{supp}(\pi) = \{v'_\pi, v''_\pi\}$ with $v'_\pi > v''_\pi$ and $\pi(\{v'_\pi\}) = \pi(\{v''_\pi\}) = \frac{1}{2}$,
2. For any $A \subseteq [\underline{v}, \overline{v}]$ measurable, $\int \mathbb{1}(v'_\pi \in A) d\overline{\mathcal{H}}(\pi) = \int \mathbb{1}(v'_{\overline{\pi}} \in A) d\mathcal{H}(\overline{\pi}|\overline{S})$ and $\int \mathbb{1}(v''_\pi \in A) d\overline{\mathcal{H}}(\pi) = \int \mathbb{1}(v'_{\underline{\pi}} \in A) d\mathcal{H}(\underline{\pi}|\underline{S})$.

Similarly, take any $\underline{\mathcal{H}} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ to be such that:

1. For any $\pi \in \mathrm{supp}(\underline{\mathcal{H}})$, $\mathrm{supp}(\pi) = \{v'_\pi, v''_\pi\}$ with $v'_\pi > v''_\pi$ and $\pi(\{v'_\pi\}) = \pi(\{v''_\pi\}) = \frac{1}{2}$,
2. For any $A \subseteq [\underline{v}, \overline{v}]$ measurable, $\int \mathbb{1}(v'_\pi \in A) d\underline{\mathcal{H}}(\pi) = \int \mathbb{1}(v''_{\overline{\pi}} \in A) d\mathcal{H}(\overline{\pi}|\overline{S})$ and $\int \mathbb{1}(v''_\pi \in A) d\underline{\mathcal{H}}(\pi) = \int \mathbb{1}(v''_{\underline{\pi}} \in A) d\mathcal{H}(\underline{\pi}|\underline{S})$.

Take any π in the support of $\overline{\mathcal{H}}$ and $\overline{\pi}$ in the support of $\mathcal{H}(\cdot|\overline{S})$ with $v'_\pi = v'_{\overline{\pi}} = v'$. By construction, $v''_\pi < 0 < v''_{\overline{\pi}}$. Hence, for any ω, $\mathbb{1}(c_{\pi^\omega} \geqslant k) \geqslant \mathbb{1}(c_{\overline{\pi}^\omega} \geqslant k)$. Moreover,

$\mathbb{1}(c_{\pi^\omega} \geqslant k) > \mathbb{1}(c_{\overline{\pi}^\omega} \geqslant k)$ for all $\omega \in (2v' - v''_{\overline{\pi}} - k, 2v' - v''_{\pi} - k)$. So:

$$\int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\gamma(\omega) > \int \mathbb{1}\left(c_{\overline{\pi}^\omega} \geqslant k\right) d\gamma(\omega),$$

and:

$$\int\int \mathbb{1}(c_{\pi^\omega} \geqslant k) d\overline{\mathcal{H}}(\pi) d\gamma(\omega) > \int\int \mathbb{1}\left(c_{\overline{\pi}^\omega} \geqslant k\right) d\mathcal{H}(\overline{\pi}|\overline{S}) d\gamma(\omega).$$

Similarly:

$$\int\int \mathbb{1}(c_{\pi^\omega} \geqslant k) d\underline{\mathcal{H}}(\pi) d\gamma(\omega) > \int\int \mathbb{1}\left(c_{\underline{\pi}^\omega} \geqslant k\right) d\mathcal{H}(\underline{\pi}|\underline{S}) d\gamma(\omega).$$

Consider the alternative plan $\hat{\mathcal{H}}$, defined as:

$$\hat{\mathcal{H}} = \mathcal{H}(\cdot|(\overline{S} \cup \underline{S})^c)(1-2p) + p\overline{\mathcal{H}} + p\underline{\mathcal{H}}.$$

It follows that:

$$\int\int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\pi) d\gamma(\omega) > \int\int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\mathcal{H}(\pi) d\gamma(\omega).$$

□

Call $\Delta_1 \subseteq \Delta([\underline{v}, \overline{v}])$ the set of district distributions $\pi \in \Delta([\underline{v}, \overline{v}])$ such that $\text{supp}(\pi) = \{v'_\pi, v''_\pi\}$, with $v'_\pi \geqslant 0$, $v''_\pi \leqslant 0$ and $\pi(\{v'_\pi\}) = \pi(\{v''_\pi\})$. Note that Δ_1 is a closed set in the weak* topology, so for any $\mathcal{H}$ optimal, $\text{supp}(\mathcal{H}) \subseteq \Delta_1$. ■

Proof of Theorem 1. The proof of this result rests on the following lemma.

Lemma 3. *There exists a bijection $t : T(\phi', \phi'') \to \Delta_2$.*

Proof. Remember $\Delta_1 \subseteq \Delta([\underline{v}, \overline{v}])$ is the set of district distributions $\pi \in \Delta([\underline{v}, \overline{v}])$ such that $\text{supp}(\pi) = \{v'_\pi, v''_\pi\}$, with $v'_\pi \geqslant 0$, $v''_\pi \leqslant 0$ and $\pi(\{v'_\pi\}) = \pi(\{v''_\pi\})$. Define the function $s : [\underline{v}, 0] \times [0, \overline{v}] \to \Delta_1$ such that, for each $(v', v'') \in [\underline{v}, 0] \times [0, \overline{v}]$, $s(v', v'') = \pi$, with $\text{supp}(\pi) = \{v', v''\}$ and $\pi(\{v'\}) = \pi(\{v''\})$. Note that s is a continuous bijection when Δ_1 is endowed with the weak* topology. Moreover, because it maps a compact set to a metrizable one, its inverse is also continuous. Thus, both s and s^{-1} map measurable sets to measurable sets.

Define the function $t : T(\phi', \phi'') \to \Delta(\Delta_1)$ such that, for all $\tau \in T(\phi', \phi'')$, for all measurable $B \subseteq \Delta_1$, $t(\tau)(B) = \tau(s^{-1}(B))$.

First, note that $t(T(\phi', \phi'')) \subseteq \Delta_2$. Indeed, take any $\tau \in T(\phi', \phi'')$. I need to show that $t(\tau)$ is feasible. That is, I show that, for all measurable $A \subseteq [\underline{v}, \overline{v}]$:

$$\int_{\Delta_1} \pi(A) dt(\tau)(\pi) = \phi(A).$$

Note that $\int_{\Delta_1} \pi(A)dt(\tau)(\pi) = \frac{1}{2}\int_{\Delta_1} \mathbb{1}(v'_\pi \in A)dt(\tau)(\pi) + \frac{1}{2}\int_{\Delta_1} \mathbb{1}(v''_\pi \in A)dt(\tau)(\pi)$. Then:

$$\int_{\Delta_1} \mathbb{1}(v'_\pi \in A)dt(\tau)(\pi) = \int_{\Delta_1} \mathbb{1}(v'_\pi \in A)d\tau(s^{-1}(\pi)) = \int_{[0,\overline{v}]\times[\underline{v},0]} \mathbb{1}(v' \in A)d\tau(v', v'').$$

Finally:

$$\int_{[0,\overline{v}]\times[\underline{v},0]} \mathbb{1}(v' \in A)d\tau(v', v'') = \left(\mathrm{proj}_{[0,\overline{v}]}\#\tau\right)(A) = \phi'(A).$$

In the same way:

$$\int_{\Delta_1} \mathbb{1}(v''_\pi \in A)dt(\tau)(\pi) = \phi''(A).$$

Similarly, $\Delta_2 \subseteq t(T(\phi', \phi''))$. Indeed, for all $\mathcal{H} \in \Delta_2$, the joint distribution τ defined as $\tau(A) = \mathcal{H}(s(A))$ for all measurable A has marginals ϕ' and ϕ''.

Finally, $\tau \neq \hat{\tau}$ implies $t(\tau) \neq t(\hat{\tau})$. Since $\tau \neq \hat{\tau}$, there exists measurable $B \subseteq [\underline{v}, 0] \times [0, \overline{v}]$ such that $\tau(B) \neq \hat{\tau}(B)$. Then:

$$t(\tau)(B) = \tau(s^{-1}(B)) \neq \hat{\tau}(s^{-1}(B)) = t(\hat{\tau})(B)$$

where the second inequality follows from the fact that s is a bijection. □

I now show that $\tau^\star \in T^\star$ if and only if $\mathcal{H}^\star = t(\tau^\star) \in \Delta_2^\star$.

For any $\mathcal{H} \in \Delta_2$, I have:

$$\begin{aligned}
&\int\int \mathbb{1}(c_{\pi^\omega} \geqslant k)d\mathcal{H}(\pi)d\gamma(\omega) \\
&= \int\int \mathbb{1}(v'_\pi - \omega \geqslant k) + \mathbb{1}(v'_\pi - \omega < k)\mathbb{1}\left(v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2}\right)d\mathcal{H}(\pi)d\gamma(\omega).
\end{aligned}$$

By switching the order of integration and further manipulating, I get:

$$\begin{aligned}
&\int\int \mathbb{1}(v'_\pi - \omega \geqslant k) + \mathbb{1}(v'_\pi - \omega < k)\mathbb{1}\left(v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2}\right)d\gamma(\omega)d\mathcal{H}(\pi) \\
&= \int\int \mathbb{1}(v'_\pi - \omega \geqslant k)d\gamma(\omega) + \int \mathbb{1}(v'_\pi - \omega < k)\mathbb{1}\left(v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2}\right)d\gamma(\omega)d\mathcal{H}(\pi) \\
&= \int G(v'_\pi - k) + G\left(2v'_\pi - v''_\pi - k\right) - G(v'_\pi - k)d\mathcal{H}(\pi) \\
&= \int G\left(2v'_\pi - v''_\pi - k\right)d\mathcal{H}(\pi).
\end{aligned}$$

Using Proposition 1, $\mathcal{H}^\star \in \Delta_2$ is optimal if and only if:

$$\int G\left(2v'_\pi - v''_\pi - k\right) d\mathcal{H}(\pi) \leqslant \int G\left(2v'_\pi - v''_\pi - k\right) d\mathcal{H}^\star(\pi) \ \text{ for all} \mathcal{H} \in \Delta_2.$$

With a change of variable:

$$\int G\left(2v'_\pi - v''_\pi - k\right) dt(\tau)(\pi) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau(s(\pi)) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau(v'_\pi, v''_\pi)$$

$$\leqslant$$

$$\int G\left(2v'_\pi - v''_\pi - k\right) dt(\tau^\star)(\pi) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau^\star(s(\pi)) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau^\star(v'_\pi, v''_\pi),$$

for all $\tau \in T(\phi', \phi'')$ and for $\tau^\star = t^{-1}(\mathcal{H}^\star)$. ■

A.2 Proofs of Section 3

In this subsection, I borrow results in Santambrogio (2015) and Chiappori et al. (2010) as building blocks to characterize the solution to (OTP). In particular, Lemma 4, Lemma 5, and Lemma 6 are adapted for my context from the above references. Given a function $f: X \to \mathbb{R}$ locally Lipschitz, call its superdifferential[20] at $x_0 \in X$, $\partial f(x_0)$. If the function is differentiable at x_0, I have that $\partial f(x_0) = \{f'(x_0)\}$. I provide the following definitions.

Definition 1. *G satisfies the twist condition whenever G is locally Lipschitz and* $\partial G(a) \cap \partial G(b) = \varnothing$ *for all* $a \neq b$.

Definition 2. *G satisfies the sub-twist condition if for all x and* $y_1 \neq y_2$, *whenever* $\partial^x G(2x - y_1 - k) \cap \partial^x G(2x - y_2 - k) \neq \varnothing$, *x is either the unique maximum or the unique minimum of* $G(2x - y_1 - k) - G(2x - y_2 - k)$.

Definition 3. *For any* $\tau \in T(\phi', \phi'')$, $\operatorname{supp}(\tau) \subseteq [0, \overline{v}] \times [\underline{v}, 0]$ *is cyclically monotone (CM) if, for every* $n \in \mathbb{N}$, *every permutation* σ, *and every finite family of points* $(v'_1, v''_1), \ldots, (v'_n, v''_n) \in \operatorname{supp}(\tau)$:

$$\sum_{i=1}^{k} G(2v'_i - v''_i - k) \geqslant \sum_{i=1}^{k} G(2v'_i - v''_{\sigma(i)} - k).$$

Then, the following lemmas hold.

[20] See, for instance, Chiappori et al. (2010).

Lemma 4. *If G satisfies the twist condition, there exists a unique, pure solution to (OTP).*

Proof. By Theorem 2 in Chiappori et al. (2010). $\square$

Lemma 5. *If G satisfies the sub-twist condition, there exists a unique solution to (OTP).*

Proof. By Theorem 3 in Chiappori et al. (2010). $\square$

Lemma 6. *If $\tau \in T(\phi', \phi'')$ is a solution to (OTP), then* $\mathrm{supp}(\tau)$ *is CM.*

Proof. By assumption, G is continuous. Hence, the result holds by Theorem 1.38 in Santambrogio (2015). $\square$

Proof of Proposition 2. Suppose G is strictly convex. Then, for all x, either $\partial G(x) = \emptyset$, or $\inf\ \partial G(x') > \sup\ \partial G(x)$ for all $x' > x$. Then, G satisfies the twist condition, and, by Lemma 4, there exists a unique, pure solution $\tau^\star$ to (OTP). Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \mathrm{supp}(\tau^\star)$, such that $v' > \tilde{v}'$. By Lemma 6, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Since G is strictly convex, the function $\Phi(v', v'') = G(2v' - v'' - k)$ is strictly submodular. Suppose that $v'' > \tilde{v}''$. By submodularity:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \leqslant \tilde{v}''$.

Suppose G is strictly concave. Then, for all x, $\sup\ \partial G(x) < \inf\ \partial G(x')$ for all $x' > x$. Then, G satisfies the twist condition, and, by Lemma 4, there exists a unique, pure solution $\tau^\star$ to (OTP). Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \mathrm{supp}(\tau^\star)$, such that $v' > \tilde{v}'$. By Lemma 6, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Since G is strictly concave, the function $\Phi(v', v'') = G(2v' - v'' - k)$ is strictly supermodular. Suppose that $v'' < \tilde{v}''$. By supermodularity:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \geqslant \tilde{v}''$.

Now, suppose G is affine. Then, for any $\tau \in T(\phi', \phi'')$:

$$\int G(2v' - v'' - k)d\tau(v', v'') = G\left(\int (2v' - v'' - k)d\tau(v', v'')\right) = G\left(\int 2v'd\phi' - \int v''d\phi'' - k\right),$$

by definition of $T(\phi', \phi'')$. Since the objective function is constant over $T(\phi', \phi'')$, I have that $T^\star = T(\phi', \phi'')$. ■

Proof of Proposition 3. First, since G is strictly convex below 0 and strictly concave above 0, it satisfies the sub-twist condition, even if it does not necessarily satisfy the twist condition. Hence, by Lemma 5, a solution τ to (OTP) exists and is unique.

Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, such that $2v' - v'' - k < 0$ and $2\tilde{v}' - \tilde{v}'' - k < 0$. Suppose that $v' > \tilde{v}'$. By Lemma 6, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Suppose that $v'' > \tilde{v}''$. Two things can happen. If $2v' - \tilde{v}'' - k \leqslant 0$ and $2\tilde{v}' - v'' - k < 0$, and since G is strictly convex below 0, I have:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

If $2v' - \tilde{v}'' - k > 0$ and $2\tilde{v}' - v'' - k < 0$, the above inequality still holds because $G(x) = 1 - G(-x)$. Indeed, I can write $G(2v' - v'' - k) - G(2\tilde{v}' - v'' - k)$ and $G(2v' - \tilde{v}'' - k) - G(2\tilde{v}' - \tilde{v}'' - k)$ as $G(2v' - v'' - k) - G(2v' - v'' - k + 2\tilde{v}' - \tilde{v}'' - k) + G(2v' - v'' - k + 2\tilde{v}' - \tilde{v}'' - k) - G(2\tilde{v}' - v'' - k)$ and $G(2v' - \tilde{v}'' - k) - G(0) + G(0) - G(2\tilde{v}' - \tilde{v}'' - k)$ respectively. By convexity of G below 0:

$$\begin{aligned} G(2v' - v'' - k) - G(2v' - v'' - k + 2\tilde{v}' - \tilde{v}'' - k) &< G(0) - G(2\tilde{v}' - \tilde{v}'' - k) \text{ and} \\ G(2v' - v'' - k + 2\tilde{v}' - \tilde{v}'' - k) - G(2\tilde{v}' - v'' - k) &< G(0) - G(-2v' + \tilde{v}'' + k) = \\ &= G(2v' - \tilde{v}'' - k) - G(0), \end{aligned}$$

where the last equality follows from $G(x) = 1 - G(-x)$. Hence, it must be $v'' \leqslant \tilde{v}''$.

Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, such that $2v' - v'' - k \geqslant 0$ and $2\tilde{v}' - \tilde{v}'' - k \geqslant 0$. Suppose that $v' > \tilde{v}'$. By Lemma 6, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Suppose that $v'' < \tilde{v}''$. Since G is strictly concave above 0, I have:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \geqslant \tilde{v}''$.

Suppose $\tau\left(\{v', v'' : 2v' - v'' - k \geqslant 0\}\right) > 0$ and $\tau\left(\{v', v'' : 2v' - v'' - k < 0\}\right) > 0$. Define the measure τ^{+} to be $\tau^{+}(\cdot) = \tau(\cdot | \{v', v'' : 2v' - v'' - k \geqslant 0\})$, and the measure τ^{-} to be $\tau^{-}(\cdot) = \tau(\cdot | \{v', v'' : 2v' - v'' - k < 0\})$. Note that τ^{+} is positive assortative, while τ^{-} is negative assortative. Moreover:

$$\tau(\cdot) = \tau\left(\{v', v'' : 2v' - v'' - k \geqslant 0\}\right)\tau^{+}(\cdot) + \tau\left(\{v', v'' : 2v' - v'' - k < 0\}\right)\tau^{-}(\cdot).$$

Suppose $\tau\left(\{v', v'' : 2v' - v'' - k \geqslant 0\}\right) = 0$. Then define $\tau^{-}(\cdot) = \tau(\cdot)$ and note that it is negative assortative.

Suppose $\tau\left(\{v', v'' : 2v' - v'' - k < 0\}\right) = 0$. Then define $\tau^{+}(\cdot) = \tau(\cdot)$ and note that it is positive assortative. ■

Proof of Proposition 4 The proof relies on the following definition and lemma.

Definition 4 (Hypo-convergence). *Consider the integral functionals* $I_{\sigma}(\tau) = \int G_{\sigma}(2v' - v'' - k)d\tau$ *and* $I(\tau) = \int \mathbb{1}(2v' - v'' - k \geqslant 0)d\tau$, *for* $\tau \in T(\phi', \phi'')$. $I_{\sigma}(\tau)$ *hypo-converges to* $I(\tau)$ *if:*

- $\forall \tau, \forall \tau_{\sigma} \to \tau$, $\limsup_{\sigma \to 0} I_{\sigma}(\tau_{\sigma}) \leqslant I(\tau)$
- $\forall \tau, \exists \tau_{\sigma} \to \tau$, $\liminf_{\sigma \to 0} I_{\sigma}(\tau_{\sigma}) \geqslant I(\tau)$.

Lemma 7. $I_{\sigma}(\tau)$ *hypo-converges to* $I(\tau)$.

Proof. First, note that:

$$\forall x, \forall x_{\sigma} \to x, \limsup_{\sigma \to 0} G_{\sigma}(x_{\sigma}) \leqslant \mathbb{1}(x \geqslant 0)$$
$$\forall x, \exists x_{\sigma} \to x, \liminf_{\sigma \to 0} G_{\sigma}(x_{\sigma}) \geqslant \mathbb{1}(x \geqslant 0).$$

Second, because G_{σ} and $\mathbb{1}(x \geqslant 0)$ are uniformly bounded, and thus uniformly integrable, I can use Theorem 2.4 in Feinberg et al. (2020) (Fatou's lemma for weakly converging measures), and have that:

$$\limsup_{\sigma \to 0} \int G_{\sigma} d\tau_{\sigma} \leqslant \int \limsup_{\sigma \to 0,\ (\hat{v}', \hat{v}'') \to (v', v'')} G_{\sigma}(2\hat{v}' - \hat{v}'' - k)d\tau \leqslant \int \mathbb{1}(2v' - v'' - k \geqslant 0)d\tau,$$

for any τ and $\tau_{\sigma} \to \tau$.

Lastly, I need to show that for any τ, there exists $\tau_\sigma \to \tau$ such that:

$$\liminf_{\sigma\to 0} \int G_\sigma d\tau_\sigma \geqslant \int \mathbb{1}(2v' - v'' - k \geqslant 0) d\tau.$$

If τ assigns no mass to $S = \{(v', v'') : 2v' - v'' - k = 0\}$, I can pick $\tau_\sigma = \tau$ and I am done. Suppose τ assigns positive mass to S. Call μ_S and ν_S the marginals of $\tau(\cdot|S)$ on $[0, \overline{v}]$ and $[\underline{v}, 0]$. For $\sigma < \frac{1}{2}$, call $\underline{\mu}_{S,\sigma}$ the distribution μ_S conditional on v' being at or below its lowest σ-quantile, and $\overline{\mu}_{S,\sigma}$ the distribution μ_S conditional on v' being at or above its highest σ-quantile. Similarly for $\underline{\nu}_{S,1-\sigma}$ and $\overline{\nu}_{S,1-\sigma}$. Call $\tau_{\sigma,<S}$ the joint distribution between $\underline{\mu}_{S,\sigma}$ and $\overline{\nu}_{S,1-\sigma}$ that matches v' and v'' in a positive assortative manner, and $\tau_{\sigma,\geqslant S}$ the joint distribution between $\overline{\mu}_{S,\sigma}$ and $\underline{\nu}_{S,1-\sigma}$ that matches v' and v'' in a positive assortative manner. For $0 < \sigma < \frac{1}{2}$, let:

$$\tau_\sigma(\cdot) = \tau(S)\sigma\tau_{\sigma,<S}(\cdot) + \tau(S)(1-\sigma)\tau_{\sigma,\geqslant S}(\cdot) + (1 - \tau(S))\tau(\cdot|S^c).$$

Since ϕ' and ϕ'' are atomless, I have that $\tau_\sigma \to \tau$ weakly as $\sigma \to 0$ and $\tau_\sigma(S) = 0$ for any $0 < \sigma < 1$. Moreover, $\tau_\sigma(\{(v', v'') : 2v' - v'' - k \geqslant 0\}) \to \tau(\{(v', v'') : 2v' - v'' - k \geqslant 0\})$ as $\sigma \to 0$. Hence, the desired inequality is satisfied and $I_\sigma(\tau)$ hypoconverges to $I(\tau)$. $\square$

For any $\sigma > 0$, call $\tau_\sigma^\star$ the unique solution to $\max_\tau I_\sigma(\tau)$. Noting that $T(\phi', \phi'')$ is compact, any limit point $\tau^\star$ of $\tau_\sigma^\star$ belongs to $T(\phi', \phi'')$ as $\sigma \to 0$. By Theorem 2.1 in Braides (2014), any such $\tau^\star$ is a solution to $\max_\tau I(\tau)$. But then it must be that $\tau^\star(\{(v', v'') : 2v' - v'' - k \geqslant 0\}) \geqslant \tau(\{(v', v'') : 2v' - v'' - k \geqslant 0\})$ for any $\tau \in T(\phi', \phi'')$. ■

Proof of Corollary 1. By Portmanteau's Theorem, for any $\delta > 0$:

$$\liminf_{\sigma\to 0} \tau_\sigma^\star(\{(v', v'') : 2v' - v'' > k - \delta\}) \geqslant \tau^\star(\{(v', v'') : 2v' - v'' > k - \delta\}) \geqslant \alpha_{max}.$$

Now, $\tau_\sigma^\star(\{(v', v'') : 2v' - v'' > k - \delta\}) = \alpha_\sigma^\star + \tau_\sigma^\star(\{(v', v'') : k - \delta < 2v' - v'' < k\})$. By Proposition 3, $\tau_\sigma^\star$ is negative assortative on $\{(v', v'') : k - \delta < 2v' - v'' < k\}$. Because ϕ is atomless, for all $\epsilon > 0$, I can find δ such that $\tau_\sigma^\star(\{(v', v'') : k - \delta < 2v' - v'' < k\}) < \epsilon$ for all σ. For instance, $\delta < \frac{\epsilon}{M}$, where M is the mode of the density of ϕ''. Moreover, if $\tau^\star(\{(v', v'') : 2v' - v'' - k = 0\}) = 0$, $\lim_{\sigma\to 0} \alpha_\sigma^\star = \alpha_{max}$. ■

A.3 Proofs of Section 4

Definition 5 (Median-Preserving Spread). *Distribution $\tilde{\phi} \in \Delta([\underline{v}, \overline{v}])$, with CDF $\tilde{F}$ and median $\tilde{v}^m$, is a median-preserving spread of distribution $\phi \in \Delta([\underline{v}, \overline{v}])$, with CDF F and median v^m, if:*

- $\tilde{v}^m = v^m$,

- $F(x) \leqslant \tilde{F}(x)$ *for* $x \leqslant v^m = \tilde{v}^m$, *and*

- $F(x) \geqslant \tilde{F}(x)$ *for* $x \geqslant v^m = \tilde{v}^m$.

Proof of Proposition 5. Note that $\tilde{\phi}'$ first-order stochastically dominates ϕ'. Then, the map $m(v') = (\tilde{F}')^{-1}(F'(v'))$ is such that $m(v') \geqslant v'$. For any $\tau \in T(\phi', \phi'')$, call $\tilde{\tau} \in T(\tilde{\phi}', \phi'')$ the distribution such that, for any measurable A:

$$\tilde{\tau}(A) = \tau(\{(v', v'') : (m(v'), v'') \in A\}).$$

I have:

$$\begin{aligned}
\max_{\tau \in T(\phi',\phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'') &= \int G\left(2v' - v'' - k\right) d\tau^\star(v', v'') \\
\leqslant \int G\left(2m(v') - v'' - k\right) d\tau^\star(v', v'') &= \int G\left(2v' - v'' - k\right) d\tilde{\tau}^\star(v', v'') \\
&\leqslant \max_{\tau \in T(\tilde{\phi}',\phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v''),
\end{aligned}$$

where the first inequality follows because G is strictly increasing. With a similar argument:

$$\max_{\tau \in T(\tilde{\phi}',\phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'') \leqslant \max_{\tau \in T(\tilde{\phi}',\tilde{\phi}'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'').$$

So:

$$\max_{\tau \in T(\phi',\phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'') \leqslant \max_{\tau \in T(\tilde{\phi}',\tilde{\phi}'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'').$$

■

Proof of Proposition 6. By Proposition 1, any district π in an optimal plan $\mathcal{H}$ must be such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$ for $v' \geqslant 0$ and $v'' \leqslant 0$. Fix shock realization ω. Consider the following cases:

- $v' - \omega \geqslant k$. Then, $c_\pi = v' - \omega \geqslant k$.

- $v' - \omega < k$ and $v' - \omega - \frac{v'' - \omega + k}{2} \geqslant 0$. Then $c_\pi = k$.

- $v' - \omega < k$ and $v' - \omega - \frac{v'' - \omega + k}{2} < 0$. Then, $c_\pi = v'' - \omega \leqslant -\omega$.

Hence, $\rho^\omega_{\mathcal{H}}((-\omega, k)) = 0$. ■

A.4 Proofs of Section 5.1

Proof of Proposition 7. First, note that a necessary, but not sufficient, condition for the designer to win district π when the realized shock is ω, is that $\pi^{\omega}(\{v : v \geqslant k\}) \geqslant \frac{2q-1}{2q}$. Suppose not. Then $\pi^{\omega}(\{v : v < k\}) > 1 - \frac{2q-1}{2q} = \frac{1}{2q}$. The equilibrium position of the Democratic candidate is $v^{q}_{\pi^{\omega},D}$, the leftmost q-quantile of π^{ω} conditional on $v < k$. Then, it must be that $\pi^{\omega}(\{v : v \leqslant v^{q}_{\pi^{\omega},D}\}) > \frac{1}{2}$, which means that the district is won by the Democrats. Now, for any redistricting plan $\mathcal{H}$ and any realization of the shock ω, define by $\chi^{\omega} \in \Delta([0,1])$ the distribution over $x^{\omega} = \pi^{\omega}(\{v : v \geqslant k\})$, that is the distribution of Republican voters across districts. Then, an upper bound on the designer utility is:

$$\int \mathbb{1}\left(x^{\omega} \geqslant \frac{2q-1}{2q}\right) d\chi^{\omega}(x^{\omega}) \leqslant \int \frac{2q}{2q-1} x^{\omega} d\chi^{\omega}(x^{\omega}) = \min\left\{1, \frac{2q}{2q-1}(1 - F(k+\omega))\right\},$$

where the equality holds by the law of iterated expectations. To finish the proof, it suffices to show that the above upper bound can always be achieved. Because F is uniform and $q \leqslant \frac{2}{3}$, it is easy to see that the upper bound is achieved by the redistricting plan that matches any $v' \in [v^{q}, \overline{v}]$ to a $v'' \in [0, v^{q}]$ and to a $v''' \in [\underline{v}, 0]$, with respective weights $\frac{2q-1}{2q}$, $\frac{1-q}{2q}$, and $\frac{1}{2}$, in a positive assortative manner. Indeed, under such assumptions $v' - v'' \leqslant v'' - v'''$ and each district is won whenever $v' - \omega \geqslant k$. ∎

Proof of Proposition 8. For any redistricting plan $\mathcal{H}$, call $H(\omega)$ the measure of districts won by the designer when the aggregate shock takes realization ω. By Proposition 7, at any realized shock ω, the designer can win at most measure $\min\left\{1, \frac{2q}{2q-1}(1 - F(k+\omega))\right\}$ of districts, so $H(\omega) \leqslant \min\left\{1, \frac{2q}{2q-1}(1 - F(k+\omega))\right\}$ at any ω. This implies that any feasible H must satisfy $H(\omega) \leqslant H^{\star}(\omega)$, where:

$$H^{\star}(\omega) = \begin{cases} 1 & if \;\; \omega \leqslant v^{q} - k \\ \frac{2q}{2q-1}(1 - F(k+\omega)) & if \;\; \omega > v^{q} - k \end{cases}$$

The designer expected utility for any feasible H is then:

$$\int H(\omega) d\gamma(\omega) \leqslant \int H^{\star}(\omega) d\gamma(\omega),$$

with strict inequality if $H(\omega) \neq H^{\star}(\omega)$ for any ω. If $H^{\star}$ is attainable at every ω, $\mathcal{H}$ is optimal if and only if it induces $H^{\star}$. $H^{\star}$ is always attainable, as shown by the proof of Proposition 7. This means that $\mathcal{H}$-almost all the districts π the designer wins if and only if the shock is at most ω must satisfy $\pi(\{v : v = k+\omega\}) = 1-\pi(\{v : v < v^q\}) = \frac{2q-1}{2q}$. However, since to win a district the designer needs at least $\frac{1}{2}$ of voters to vote for the Republican candidate and $\frac{2q-1}{q} \leqslant \frac{1}{2}$, some voters with $v < v^q$ must vote for R. Precisely $\frac{1}{2} - \frac{2q-1}{2q}$ additional voters must prefer $v = k + \omega$, the Republican candidate, to $v^q_{\pi^\omega,D}$, the Democratic candidate, with $v^q_{\pi^\omega,D}$ being the lowest q - quantile of π^ω conditional on $v < k$. But given that $\pi(\{v : v < k\}) = \frac{1}{2q}$, the Democratic candidate sits at the lowest median of π. For him not to win the district, it must be that:

$$|v^q_{\pi^\omega,D} - v^m_{\pi^\omega}| \geqslant |v^m_{\pi^\omega} - (k+\omega)|.$$

■

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Supplemental Appendix

B A Model of Two-Stage District Elections

In this appendix, I develop a model of probabilistic voting at the district level, providing a foundation for the dependence of district candidates' positions on the conditional medians of their respective vote base. While equilibrium policy divergence can be supported under a variety of models, the dependence of such policies on conditional medians is typical of two-stage election mechanisms. I propose one such model in which, like in many others, two forces work in opposing directions to determine equilibrium policies. One such force, a centripetal force, drives positions towards the district median, fueled by the concern voters have with the ability of their first-stage candidate to win general elections. Another force, a centrifugal force, moves positions away from each other, driven by first-stage voters' uncertainty about the exact position of the district median. Similarly to what was suggested, among others, by Owen and Grofman (2006), equilibrium policies tend to be driven towards party medians.

Consider a continuum of voters in a district. Each voter i has policy preferences given by single-peaked utility $u(\cdot, v_i)$, with $v_i \in [\underline{v}, \overline{v}]$ being the voter's unique ideal point. Further, assume that the distribution π of v_i admits a unique median, conditional median given $v_i \geqslant k$, and conditional median given $v_i < k$, denoted by v^m_π, $v^m_{\pi,R}$, and $v^m_{\pi,D}$.

Two-stage elections are held in the district. In the first, primaries stage, there are two Republican and two Democratic candidates. Voters with $v_i \geqslant k$ elect one of the Republican candidates, while voters with $v_i < k$ elect one of the Democratic candidates, by simple majority, with ties broken uniformly at random. In the second stage, general elections, all voters elect a district representative among the two first-stage winners, by simple majority, with ties broken uniformly at random.

Before the first-stage elections, candidates simultaneously announce a policy position and commit to it. They receive a payoff of 1 if they win the second-stage elections and 0 otherwise.

Suppose that voters and candidates do not know the position of the overall district median v^m_π, and believe it is distributed according to $H \in \Delta([\underline{v}, \overline{v}])$. In addition, candidates know the position of the conditional medians $v^m_{\pi,R}$ and $v^m_{\pi,D}$.

Given voters' single-peaked preferences, the winner of the general elections is the candidate closer to the district median. In the first stage, voters best respond to the anticipated position of the opposing nominee by voting for the candidate maximizing their expected utility. More formally, given the position of the opposing nominee y, a voter with ideal point t votes for the candidate whose position maximizes the following:

$$U(x;y,t) = u(x,t)p(x,y) + u(y,t)(1 - p(x,y)),$$

where $p(x,y)$ is the probability of x winning against y in general elections. As already stated, equilibrium policies in such a model are driven by party medians, but their exact value depends on the functional form of u and H. I suggest a *linear setting*, where voter preferences are linear and uncertainty is uniform. I prove the following.

Proposition 9. *Suppose $u(x,v_i) = -|x - v_i|$ and H is uniform on $[\underline{v},\overline{v}]$. There exists a unique Nash equilibrium where the Republican and Democratic candidates set positions $c_{\pi,R} = v^m_{\pi,R}$ and $c_{\pi,D} = v^m_{\pi,D}$, respectively.*

The advantage of the linear setting is that voters' expected utilities in the primaries turn out to be single-peaked, with maximum reached at each voter's ideal point. Then, it can be argued that any candidate positioning at the party median will win first-stage elections against any other candidate at a different position, irrespective of the opposing party's behavior. Alternatively, Owen and Grofman (2006) consider a model where $u(x,v_i) = e^{-\alpha|x-v_i|}$, for some parameter $\alpha > 0$, and uncertainty is normal around the median, with standard deviation σ. In their model, voters' expected utility during primary elections is not necessarily single-peaked. Hence, they need to explicitly rule out a particular strategic behavior in which some first-stage voters anticipate that they will prefer the opponents' candidate at general elections and purposefully sabotage their own primaries by voting for an extremist[21]. Under this assumption, they prove the following:

Proposition 10. (Owen and Grofman, 2006). If $\sigma \geqslant \frac{\max\left\{1-e^{-\alpha(v^m_\pi - v^m_{\pi,D})}, 1-e^{-\alpha(v^m_{\pi,R} - v^m_\pi)}\right\}}{\alpha\sqrt{2\pi}}$,

[21] Under uniform uncertainty, extreme events have a sufficiently high probability of happening, so that voters are deterred from this kind of "political gamble". Indeed, if they sabotage primaries by voting for an extremist, they run the risk of such a candidate winning general elections as well.

there exists a unique Nash equilibrium where the Republican and Democratic candidates set positions $c_{\pi,R} = v^m_{\pi,R}$ and $c_{\pi,D} = v^m_{\pi,D}$, respectively.

C Discussion on Technical Assumptions

Throughout the paper, two modeling assumptions play a role in the formal analysis: (i) the tie-breaking rules for the median v^m_π and conditional medians $v^m_{\pi,D}$ and $v^m_{\pi,R}$, and (ii) the assumption that, in a district π with $\text{supp}(\pi) \subseteq (-\infty, k]$ (respectively, $\text{supp}(\pi) \subseteq [k, \infty)$), the Republican candidate $c_{\pi,R}$ (respectively, Democratic candidate $c_{\pi,D}$) defaults to position k. I now briefly discuss the motivation for these choices and their implications.

Given the tie-breaking rule for v^m_π, the one for the conditional median $v^m_{\pi,R}$ and the assumption that, in a district with $\text{supp}(\pi) \subseteq [k, \infty)$, the Democratic candidate defaults to position k, do not play a substantive role in the analysis. I adopt them for coherence with the other assumptions in the model.

I can now turn to the remaining assumptions: the tie-breaking rules for v^m_π and $v^m_{\pi,D}$ and the assumption that the Republican candidate defaults to k in the absence of Republican voters. In the model without aggregate uncertainty, these assumptions are needed to guarantee the existence of an optimum and can be derived from relatively simple limit arguments. To illustrate how, suppose that for a candidate to win either the primary or a general election, they need a strict majority of votes–say, by a margin of at least $\frac{\epsilon-2\delta}{1-\epsilon}$ and δ, respectively, for some small values $\delta, \epsilon > 0$, with $\delta < \frac{\epsilon}{2}$. I can construct a sequence of redistricting plans indexed by ϵ and δ that converge to our solution as we bring these parameters to zero. Given (ϵ, δ), partition the support of F into four intervals: $[\underline{v},\ F^{-1}\left(\frac{1}{2}-\delta\right)]$, $[F^{-1}\left(\frac{1}{2}-\delta\right), k]$, $[k, F^{-1}(F(k)+\epsilon)]$, $[F^{-1}(F(k)+\epsilon), \overline{v}]$.[22] There are two classes of districts.

The first class of districts is made by pairing types v' above $F^{-1}\left(\frac{1}{2}-\delta\right)$ with mass $\frac{1}{2}+\delta$ and types v'' below $F^{-1}\left(\frac{1}{2}-\delta\right)$ with mass $\frac{1}{2}-\delta$. If v' is above k the district is won by a δ margin. Otherwise, it is lost irrespective of the position taken by the Republican candidate.

The second class of districts is made of at least three types of voters: a type $u' \in$

[22] Here I consider the case where $k > 0$. The opposite case is trivial, as the designer has a majority of supporters in the population.

$[F^{-1}(\frac{1}{2}-\delta), k]$ with mass $\frac{1}{2}+\delta-\epsilon$, a type u'' below $F^{-1}(\frac{1}{2}-\delta)$ with mass $\frac{1}{2}-\delta$, and mass ϵ of types in $[k, F^{-1}(F(k)+\epsilon)]$. Note that u' is the median of the district, u'' is the Democratic candidate, who wins primaries, and the Republican candidate is some $u''' \in [k, F^{-1}(F(k)+\epsilon)]$, who wins the general election by a δ margin if $u'''-u' \leqslant u'-u''$. Districts in this class are created if and only if they would be won. How many can be created depends on F. Having Republican types as close as possible to k ensures that as many as possible are created. Then, the discussed assumptions appear as limit points as δ and ϵ go to zero, keeping $\delta < \frac{\epsilon}{2}$.

Note that the possibility that a district contains no Republican voters does not constitute an issue in the constructed sequence. In fact, in the model without aggregate uncertainty, no issue of well-definedness of the position of candidates in the absence of a party base arises. This is no longer the case in the presence of an aggregate shock. Even if the designer allocates an arbitrarily small fraction of Republicans close to k to a district composed mostly of Democrats, a sufficiently large realization of the shock can shift this fraction below k, giving rise to the question of where the Republican candidate would locate. To maintain coherence with the case of no aggregate uncertainty, I therefore preserve the same assumptions that guarantee existence without aggregate uncertainty. This can be interpreted as assuming that whenever the entire support of a district falls below k, an arbitrarily small upper tail of the district remains at k, insensitive to the shock. This tail moves with the shock again only when the rest of the district rises above k (similarly, one can define what happens when a district falls entirely above k).[23]

Moreover, this assumption has the advantage of preserving the continuity of candidates' positions with respect to the shock; if, for instance, we assumed that a district whose support falls entirely below k is automatically won by the Democrats without even a Republican candidate, this would generate an unappealing discontinuity of candidates position with respect to the shock.

[23]Note that this assumption would not help in the extension of Section 5.1, where a non-vanishing fraction of Republicans is needed to win a district.

D Proofs of Appendix B

Proof of Proposition 9. Consider a Democratic voter with ideal point t (the reasoning is similar for a Republican voter). Given position y of the Republican candidate, the expected utility of electing a first-stage Democratic candidate with position x is:

$$U(x;y,t) = u(x,t)p(x,y) + u(y,t)(1-p(x,y)),$$

where $p(x,y)$ is the probability of the Democratic candidate winning against the Republican candidate in the general elections. The proof of this result relies on the following lemma.

Lemma 8. *$U(x;y,t)$ is strictly increasing for $x < t$, strictly decreasing for $x > t$, and achieves its maximum at $x = t$.*

Proof. First, suppose $y > t$. Consider three cases:

Case $x > y$. First, note that $U(x;y,t) \leqslant U(y;y,t)$. Moreover, $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (y-t)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} - \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -1 + \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{x-y}{2(\overline{v} - \underline{v})} = -1 + \frac{x-2\underline{v}}{2(\overline{v} - \underline{v})} < 0$$

Case $t < x < y$. $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (y-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -\frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y-x}{2(\overline{v} - \underline{v})} = \frac{-2x+2\underline{v}}{2(\overline{v} - \underline{v})} < 0$$

Case $x < t$. $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) - (y-t)\left(1-\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) + \frac{x-t}{2(\overline{v}-\underline{v})} + \frac{y-t}{2(\overline{v}-\underline{v})} =$$

$$= \frac{x+y-2\underline{v}}{2(\overline{v}-\underline{v})} + \frac{y+x-2t}{2(\overline{v}-\underline{v})} = \frac{2(x+y-t-\underline{v})}{2(\overline{v}-\underline{v})} > 0$$

Second, suppose $y < t$. Consider three cases:

Case $x < y$. First, note that $U(x;y,t) < U(y;y,t)$. Moreover, $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) - (t-y)\left(1-\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) + \frac{x-t}{2(\overline{v}-\underline{v})} - \frac{y-t}{2(\overline{v}-\underline{v})} =$$

$$= \frac{x+y-2\underline{v}}{2(\overline{v}-\underline{v})} + \frac{x-y}{2(\overline{v}-\underline{v})} = \frac{x-2\underline{v}}{2(\overline{v}-\underline{v})} > 0$$

Case $y < x < t$. $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(1-\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) - (t-y)\left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(1-\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) - \frac{x-t}{2(\overline{v}-\underline{v})} + \frac{y-t}{2(\overline{v}-\underline{v})} =$$

$$= 1-\frac{x+y-2\underline{v}}{2(\overline{v}-\underline{v})} + \frac{y-x}{2(\overline{v}-\underline{v})} = 1-\frac{-2x+2\underline{v}}{2(\overline{v}-\underline{v})} > 0$$

Case $x > t$. $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(1-\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right) - (t-y)\left(\frac{\frac{x+y}{2}-\underline{v}}{\overline{v}-\underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -1 + \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y+x-2t}{2(\overline{v} - \underline{v})} = -1 + \frac{2(x+y-t-\underline{v})}{2(\overline{v} - \underline{v})} < 0.$$

Finally, note that $U(x;y,t)$ is continuous, so it is reaches its maximum at $x = t$, it is strictly decreasing for $x > t$, and strictly increasing for $x < t$.

□

Given Lemma 8, a Democratic candidate at $v^m_{\pi,D}$ will win first-stage elections against any candidate at a different position (and similarly for Republicans). Indeed, Democratic voters to the right of $v^m_{\pi,D}$, accounting for half of all Democratic voters, prefer a candidate at $v^m_{\pi,D}$ to any other candidate to the left of $v^m_{\pi,D}$. Moreover, Democratic voters to the left of $v^m_{\pi,D}$, prefer a candidate at $v^m_{\pi,D}$ to any other candidate to the right of $v^m_{\pi,D}$. Importantly, this reasoning is independent of the Republican candidate's position y. Since positioning at $v^m_{\pi,D}$ ($v^m_{\pi,R}$, respectively) gives a Democratic (Republican) first-stage candidate a positive probability of winning second-stage elections, there exists an equilibrium where both Democratic candidates set at $v^m_{\pi,D}$ and both Republican candidates set at $v^m_{\pi,R}$.

Now, I show there can not be any other equilibrium. First, any situation where first-stage candidates do not tie can not be an equilibrium, because the losing candidate can move to $v^m_{\pi,D}$ ($v^m_{\pi,R}$) and have positive probability of winning second-statge elections. Second, any situation where first-stage candidates do not set the same position can not be an equilibrium. To see this, note that, in order to have different positions and tie at the same time, one of the Democratic (Republican) first-stage candidate needs to choose a position to the left (right) of $v^m_{\pi,D}$ ($v^m_{\pi,R}$). However, any such position is outside of the support of the median distribution, since candidates know the position of the conditional medians, and is therefore dominated by $v^m_{\pi,D}$ ($v^m_{\pi,R}$). Finally, suppose the two first-stage candidates set at the same position, different from $v^m_{\pi,D}$ ($v^m_{\pi,R}$). Then, there exists small $\epsilon > 0$ such that one of the candidates has a profitable deviation by moving closer to $v^m_{\pi,D}$ ($v^m_{\pi,R}$) by ϵ.

■

E Extra Figures

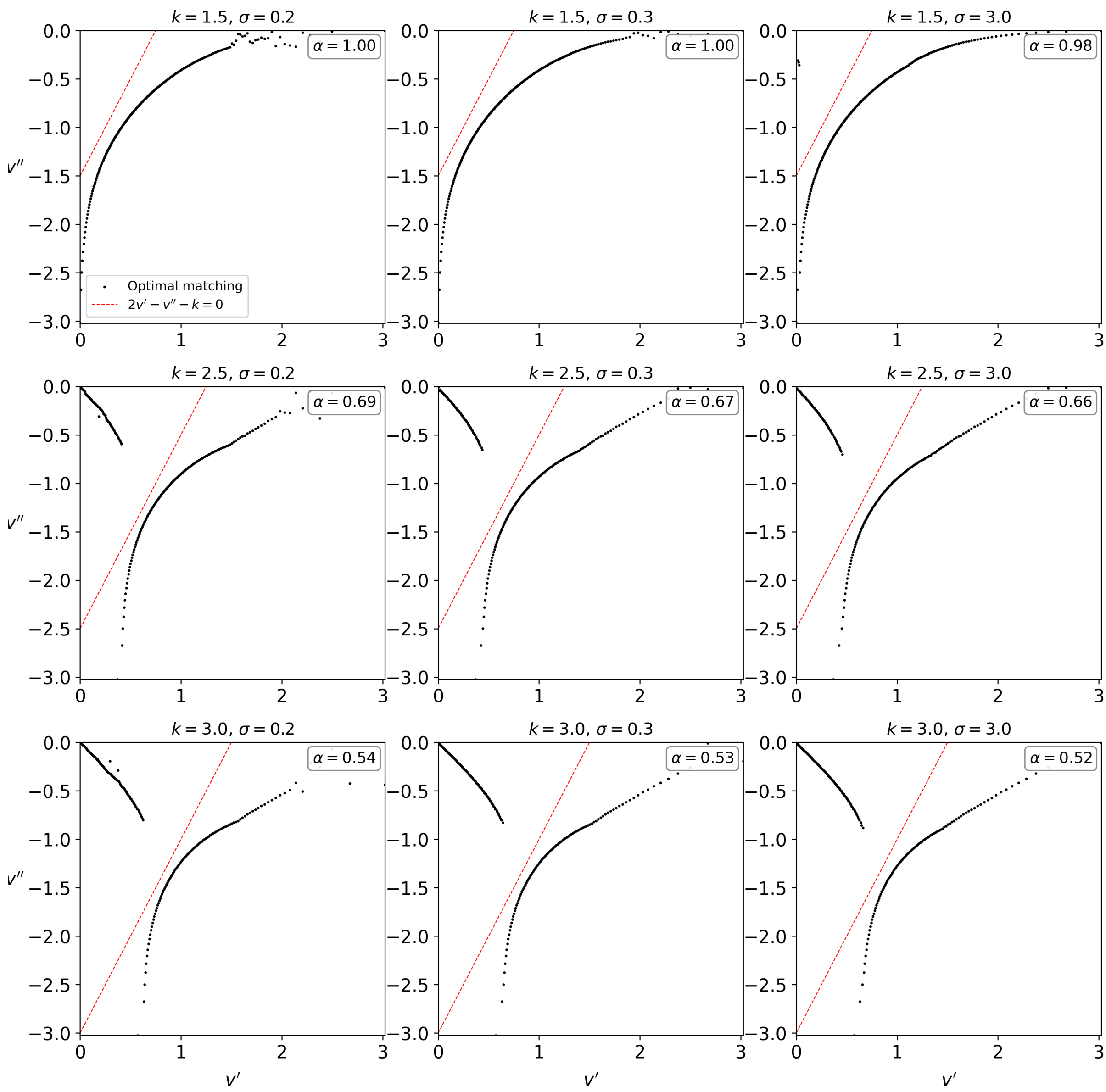

Figure 13: Simulated solutions as the variance σ of the normal shock and k change, for F normal with variance 1.

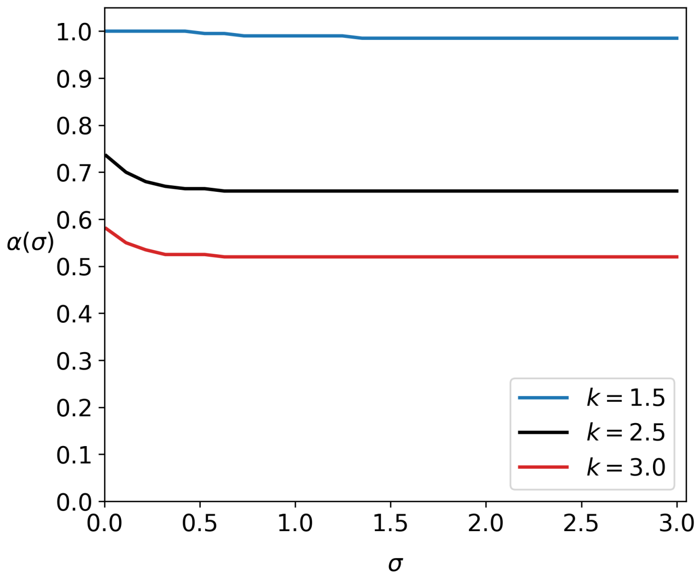

Figure 14: Simulated values of α as the variance σ of the normal shock changes, for F normal with variance 1.